



1062CH14

STATISTICS 13

13.1 Introduction

In Class IX, you have studied the classification of given data into ungrouped as well as grouped frequency distributions. You have also learnt to represent the data pictorially in the form of various graphs such as bar graphs, histograms (including those of varying widths) and frequency polygons. In fact, you went a step further by studying certain numerical representatives of the ungrouped data, also called measures of central tendency, namely, *mean*, *median* and *mode*. In this chapter, we shall extend the study of these three measures, i.e., mean, median and mode from ungrouped data to that of *grouped data*. We shall also discuss the concept of cumulative frequency, the cumulative frequency distribution and how to draw cumulative frequency curves, called *ogives*.

13.2 Mean of Grouped Data

The mean (or average) of observations, as we know, is the sum of the values of all the observations divided by the total number of observations. From Class IX, recall that if $x_1, x_2, \dots, x_n$ are observations with respective frequencies $f_1, f_2, \dots, f_n$, then this means observation x_1 occurs f_1 times, x_2 occurs f_2 times, and so on.

Now, the sum of the values of all the observations = $f_1x_1 + f_2x_2 + \dots + f_nx_n$, and the number of observations = $f_1 + f_2 + \dots + f_n$.

So, the mean $\bar{x}$ of the data is given by

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + \dots + f_nx_n}{f_1 + f_2 + \dots + f_n}$$

Recall that we can write this in short form by using the Greek letter Σ (capital sigma) which means summation. That is,

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

which, more briefly, is written as $\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i}$, if it is understood that i varies from 1 to n .

Let us apply this formula to find the mean in the following example.

Example 1 : The marks obtained by 30 students of Class X of a certain school in a Mathematics paper consisting of 100 marks are presented in table below. Find the mean of the marks obtained by the students.

Marks obtained (x_i)	10	20	36	40	50	56	60	70	72	80	88	92	95
Number of students (f_i)	1	1	3	4	3	2	4	4	1	1	2	3	1

Solution: Recall that to find the mean marks, we require the product of each x_i with the corresponding frequency f_i . So, let us put them in a column as shown in Table 13.1.

Table 13.1

Marks obtained (x_i)	Number of students (f_i)	$f_i x_i$
10	1	10
20	1	20
36	3	108
40	4	160
50	3	150
56	2	112
60	4	240
70	4	280
72	1	72
80	1	80
88	2	176
92	3	276
95	1	95
Total	$\Sigma f_i = 30$	$\Sigma f_i x_i = 1779$

Now,

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{1779}{30} = 59.3$$

Therefore, the mean marks obtained is 59.3.

In most of our real life situations, data is usually so large that to make a meaningful study it needs to be condensed as grouped data. So, we need to convert given ungrouped data into grouped data and devise some method to find its mean.

Let us convert the ungrouped data of Example 1 into grouped data by forming class-intervals of width, say 15. Remember that, while allocating frequencies to each class-interval, students falling in any upper class-limit would be considered in the next class, e.g., 4 students who have obtained 40 marks would be considered in the class-interval 40-55 and not in 25-40. With this convention in our mind, let us form a grouped frequency distribution table (see Table 13.2).

Table 13.2

Class interval	10 - 25	25 - 40	40 - 55	55 - 70	70 - 85	85 - 100
Number of students	2	3	7	6	6	6

Now, for each class-interval, we require a point which would serve as the representative of the whole class. *It is assumed that the frequency of each class-interval is centred around its mid-point.* So the *mid-point* (or *class mark*) of each class can be chosen to represent the observations falling in the class. Recall that we find the mid-point of a class (or its class mark) by finding the average of its upper and lower limits. That is,

$$\text{Class mark} = \frac{\text{Upper class limit} + \text{Lower class limit}}{2}$$

With reference to Table 13.2, for the class 10-25, the class mark is $\frac{10+25}{2}$, i.e.,

17.5. Similarly, we can find the class marks of the remaining class intervals. We put them in Table 13.3. These class marks serve as our x_i 's. Now, in general, for the i th class interval, we have the frequency f_i corresponding to the class mark x_i . We can now proceed to compute the mean in the same manner as in Example 1.

Table 13.3

Class interval	Number of students (f_i)	Class mark (x_i)	$f_i x_i$
10 - 25	2	17.5	35.0
25 - 40	3	32.5	97.5
40 - 55	7	47.5	332.5
55 - 70	6	62.5	375.0
70 - 85	6	77.5	465.0
85 - 100	6	92.5	555.0
Total	$\Sigma f_i = 30$		$\Sigma f_i x_i = 1860.0$

The sum of the values in the last column gives us $\Sigma f_i x_i$. So, the mean $\bar{x}$ of the given data is given by

$$\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{1860.0}{30} = 62$$

This new method of finding the mean is known as the **Direct Method**.

We observe that Tables 13.1 and 13.3 are using the same data and employing the same formula for the calculation of the mean but the results obtained are different. Can you think why this is so, and which one is more accurate? The difference in the two values is because of the mid-point assumption in Table 13.3, 59.3 being the exact mean, while 62 an approximate mean.

Sometimes when the numerical values of x_i and f_i are large, finding the product of x_i and f_i becomes tedious and time consuming. So, for such situations, let us think of a method of reducing these calculations.

We can do nothing with the f_i 's, but we can change each x_i to a smaller number so that our calculations become easy. How do we do this? What about subtracting a fixed number from each of these x_i 's? Let us try this method.

The first step is to choose one among the x_i 's as the *assumed mean*, and denote it by ' a '. Also, to further reduce our calculation work, we may take ' a ' to be that x_i which lies in the centre of $x_1, x_2, \dots, x_n$. So, we can choose $a = 47.5$ or $a = 62.5$. Let us choose $a = 47.5$.

The next step is to find the difference d_i between a and each of the x_i 's, that is, the **deviation** of ' a ' from each of the x_i 's.

i.e.,
$$d_i = x_i - a = x_i - 47.5$$

The third step is to find the product of d_i with the corresponding f_i , and take the sum of all the $f_i d_i$'s. The calculations are shown in Table 13.4.

Table 13.4

Class interval	Number of students (f_i)	Class mark (x_i)	$d_i = x_i - 47.5$	$f_i d_i$
10 - 25	2	17.5	-30	-60
25 - 40	3	32.5	-15	-45
40 - 55	7	47.5	0	0
55 - 70	6	62.5	15	90
70 - 85	6	77.5	30	180
85 - 100	6	92.5	45	270
Total	$\Sigma f_i = 30$			$\Sigma f_i d_i = 435$

So, from Table 13.4, the mean of the deviations, $\bar{d} = \frac{\Sigma f_i d_i}{\Sigma f_i}$.

Now, let us find the relation between $\bar{d}$ and $\bar{x}$.

Since in obtaining d_i , we subtracted 'a' from each x_i , so, in order to get the mean $\bar{x}$, we need to add 'a' to $\bar{d}$. This can be explained mathematically as:

$$\text{Mean of deviations, } \bar{d} = \frac{\Sigma f_i d_i}{\Sigma f_i}$$

$$\begin{aligned} \text{So, } \bar{d} &= \frac{\Sigma f_i (x_i - a)}{\Sigma f_i} \\ &= \frac{\Sigma f_i x_i}{\Sigma f_i} - \frac{\Sigma f_i a}{\Sigma f_i} \\ &= \bar{x} - a \frac{\Sigma f_i}{\Sigma f_i} \end{aligned}$$

$$= \bar{x} - a$$

$$\text{So, } \bar{x} = a + \bar{d}$$

$$\text{i.e., } \bar{x} = a + \frac{\Sigma f_i d_i}{\Sigma f_i}$$

Substituting the values of a , $\Sigma f_i d_i$ and Σf_i from Table 13.4, we get

$$\bar{x} = 47.5 + \frac{435}{30} = 47.5 + 14.5 = 62.$$

Therefore, the mean of the marks obtained by the students is 62.

The method discussed above is called the **Assumed Mean Method**.

Activity 1 : From the Table 13.3 find the mean by taking each of x_i (i.e., 17.5, 32.5, and so on) as 'a'. What do you observe? You will find that the mean determined in each case is the same, i.e., 62. (Why ?)

So, we can say that the value of the mean obtained does not depend on the choice of 'a'.

Observe that in Table 13.4, the values in Column 4 are all multiples of 15. So, if we divide the values in the entire Column 4 by 15, we would get smaller numbers to multiply with f_i . (Here, 15 is the class size of each class interval.)

So, let $u_i = \frac{x_i - a}{h}$, where a is the assumed mean and h is the class size.

Now, we calculate u_i in this way and continue as before (i.e., find $f_i u_i$ and then $\Sigma f_i u_i$). Taking $h = 15$, let us form Table 13.5.

Table 13.5

Class interval	f_i	x_i	$d_i = x_i - a$	$u_i = \frac{x_i - a}{h}$	$f_i u_i$
10 - 25	2	17.5	-30	-2	-4
25 - 40	3	32.5	-15	-1	-3
40 - 55	7	47.5	0	0	0
55 - 70	6	62.5	15	1	6
70 - 85	6	77.5	30	2	12
85 - 100	6	92.5	45	3	18
Total	$\Sigma f_i = 30$				$\Sigma f_i u_i = 29$

Let

$$\bar{u} = \frac{\Sigma f_i u_i}{\Sigma f_i}$$

Here, again let us find the relation between $\bar{u}$ and $\bar{x}$.

We have,

$$u_i = \frac{x_i - a}{h}$$

Therefore,

$$\begin{aligned}\bar{u} &= \frac{\sum f_i \frac{(x_i - a)}{h}}{\sum f_i} = \frac{1}{h} \left[\frac{\sum f_i x_i - a \sum f_i}{\sum f_i} \right] \\ &= \frac{1}{h} \left[\frac{\sum f_i x_i}{\sum f_i} - a \frac{\sum f_i}{\sum f_i} \right] \\ &= \frac{1}{h} [\bar{x} - a]\end{aligned}$$

So,

$$h\bar{u} = \bar{x} - a$$

i.e.,

$$\bar{x} = a + h\bar{u}$$

So,

$$\bar{x} = a + h \left(\frac{\sum f_i u_i}{\sum f_i} \right)$$

Now, substituting the values of a , h , $\sum f_i u_i$ and $\sum f_i$ from Table 14.5, we get

$$\begin{aligned}\bar{x} &= 47.5 + 15 \times \left(\frac{29}{30} \right) \\ &= 47.5 + 14.5 = 62\end{aligned}$$

So, the mean marks obtained by a student is 62.

The method discussed above is called the **Step-deviation** method.

We note that :

- the step-deviation method will be convenient to apply if all the d_i 's have a common factor.
- The mean obtained by all the three methods is the same.
- The assumed mean method and step-deviation method are just simplified forms of the direct method.
- The formula $\bar{x} = a + h\bar{u}$ still holds if a and h are not as given above, but are any non-zero numbers such that $u_i = \frac{x_i - a}{h}$.

Let us apply these methods in another example.

Example 2 : The table below gives the percentage distribution of female teachers in the primary schools of rural areas of various states and union territories (U.T.) of India. Find the mean percentage of female teachers by all the three methods discussed in this section.

Percentage of female teachers	15 - 25	25 - 35	35 - 45	45 - 55	55 - 65	65 - 75	75 - 85
Number of States/U.T.	6	11	7	4	4	2	1

Source : *Seventh All India School Education Survey conducted by NCERT*

Solution : Let us find the class marks, x_i , of each class, and put them in a column (see Table 13.6):

Table 13.6

Percentage of female teachers	Number of States /U.T. (f_i)	x_i
15 - 25	6	20
25 - 35	11	30
35 - 45	7	40
45 - 55	4	50
55 - 65	4	60
65 - 75	2	70
75 - 85	1	80

Here we take $a = 50$, $h = 10$, then $d_i = x_i - 50$ and $u_i = \frac{x_i - 50}{10}$.

We now find d_i and u_i and put them in Table 13.7.

Table 13.7

Percentage of female teachers	Number of states/U.T. (f_i)	x_i	$d_i = x_i - 50$	$u_i = \frac{x_i - 50}{10}$	$f_i x_i$	$f_i d_i$	$f_i u_i$
15 - 25	6	20	-30	-3	120	-180	-18
25 - 35	11	30	-20	-2	330	-220	-22
35 - 45	7	40	-10	-1	280	-70	-7
45 - 55	4	50	0	0	200	0	0
55 - 65	4	60	10	1	240	40	4
65 - 75	2	70	20	2	140	40	4
75 - 85	1	80	30	3	80	30	3
Total	35				1390	-360	-36

From the table above, we obtain $\Sigma f_i = 35$, $\Sigma f_i x_i = 1390$,

$$\Sigma f_i d_i = -360, \quad \Sigma f_i u_i = -36.$$

Using the direct method, $\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{1390}{35} = 39.71$

Using the assumed mean method,

$$\bar{x} = a + \frac{\Sigma f_i d_i}{\Sigma f_i} = 50 + \frac{(-360)}{35} = 39.71$$

Using the step-deviation method,

$$\bar{x} = a + \left(\frac{\Sigma f_i u_i}{\Sigma f_i} \right) \times h = 50 + \left(\frac{-36}{35} \right) \times 10 = 39.71$$

Therefore, the mean percentage of female teachers in the primary schools of rural areas is 39.71.

Remark : The result obtained by all the three methods is the same. So the choice of method to be used depends on the numerical values of x_i and f_i . If x_i and f_i are sufficiently small, then the direct method is an appropriate choice. If x_i and f_i are numerically large numbers, then we can go for the assumed mean method or step-deviation method. If the class sizes are unequal, and x_i are large numerically, we can still apply the step-deviation method by taking h to be a suitable divisor of all the d_i 's.

Example 3 : The distribution below shows the number of wickets taken by bowlers in one-day cricket matches. Find the mean number of wickets by choosing a suitable method. What does the mean signify?

Number of wickets	20 - 60	60 - 100	100 - 150	150 - 250	250 - 350	350 - 450
Number of bowlers	7	5	16	12	2	3

Solution : Here, the class size varies, and the x_i s are large. Let us still apply the step-deviation method with $a = 200$ and $h = 20$. Then, we obtain the data as in Table 13.8.

Table 13.8

Number of wickets taken	Number of bowlers (f_i)	x_i	$d_i = x_i - 200$	$u_i = \frac{d_i}{20}$	$u_i f_i$
20 - 60	7	40	-160	-8	-56
60 - 100	5	80	-120	-6	-30
100 - 150	16	125	-75	-3.75	-60
150 - 250	12	200	0	0	0
250 - 350	2	300	100	5	10
350 - 450	3	400	200	10	30
Total	45				-106

So, $\bar{u} = \frac{-106}{45}$. Therefore, $\bar{x} = 200 + 20\left(\frac{-106}{45}\right) = 200 - 47.11 = 152.89$.

This tells us that, on an average, the number of wickets taken by these 45 bowlers in one-day cricket is 152.89.

Now, let us see how well you can apply the concepts discussed in this section!

Activity 2 :

Divide the students of your class into three groups and ask each group to do one of the following activities.

1. Collect the marks obtained by all the students of your class in Mathematics in the latest examination conducted by your school. Form a grouped frequency distribution of the data obtained.
2. Collect the daily maximum temperatures recorded for a period of 30 days in your city. Present this data as a grouped frequency table.
3. Measure the heights of all the students of your class (in cm) and form a grouped frequency distribution table of this data.

After all the groups have collected the data and formed grouped frequency distribution tables, the groups should find the mean in each case by the method which they find appropriate.

EXERCISE 13.1

1. A survey was conducted by a group of students as a part of their environment awareness programme, in which they collected the following data regarding the number of plants in 20 houses in a locality. Find the mean number of plants per house.

Number of plants	0 - 2	2 - 4	4 - 6	6 - 8	8 - 10	10 - 12	12 - 14
Number of houses	1	2	1	5	6	2	3

Which method did you use for finding the mean, and why?

2. Consider the following distribution of daily wages of 50 workers of a factory.

Daily wages (in ₹)	500 - 520	520 - 540	540 - 560	560 - 580	580 - 600
Number of workers	12	14	8	6	10

Find the mean daily wages of the workers of the factory by using an appropriate method.

3. The following distribution shows the daily pocket allowance of children of a locality. The mean pocket allowance is Rs 18. Find the missing frequency f .

Daily pocket allowance (in ₹)	11 - 13	13 - 15	15 - 17	17 - 19	19 - 21	21 - 23	23 - 25
Number of children	7	6	9	13	f	5	4

4. Thirty women were examined in a hospital by a doctor and the number of heartbeats per minute were recorded and summarised as follows. Find the mean heartbeats per minute for these women, choosing a suitable method.

Number of heartbeats per minute	65 - 68	68 - 71	71 - 74	74 - 77	77 - 80	80 - 83	83 - 86
Number of women	2	4	3	8	7	4	2

5. In a retail market, fruit vendors were selling mangoes kept in packing boxes. These boxes contained varying number of mangoes. The following was the distribution of mangoes according to the number of boxes.

Number of mangoes	50 - 52	53 - 55	56 - 58	59 - 61	62 - 64
Number of boxes	15	110	135	115	25

Find the mean number of mangoes kept in a packing box. Which method of finding the mean did you choose?

6. The table below shows the daily expenditure on food of 25 households in a locality.

Daily expenditure (in ₹)	100 - 150	150 - 200	200 - 250	250 - 300	300 - 350
Number of households	4	5	12	2	2

Find the mean daily expenditure on food by a suitable method.

7. To find out the concentration of SO_2 in the air (in parts per million, i.e., ppm), the data was collected for 30 localities in a certain city and is presented below:

Concentration of SO_2 (in ppm)	Frequency
0.00 - 0.04	4
0.04 - 0.08	9
0.08 - 0.12	9
0.12 - 0.16	2
0.16 - 0.20	4
0.20 - 0.24	2

Find the mean concentration of SO_2 in the air.

8. A class teacher has the following absentee record of 40 students of a class for the whole term. Find the mean number of days a student was absent.

Number of days	0 - 6	6 - 10	10 - 14	14 - 20	20 - 28	28 - 38	38 - 40
Number of students	11	10	7	4	4	3	1

9. The following table gives the literacy rate (in percentage) of 35 cities. Find the mean literacy rate.

Literacy rate (in %)	45 - 55	55 - 65	65 - 75	75 - 85	85 - 95
Number of cities	3	10	11	8	3

13.3 Mode of Grouped Data

Recall from Class IX, a mode is that value among the observations which occurs most often, that is, the value of the observation having the maximum frequency. Further, we discussed finding the mode of ungrouped data. Here, we shall discuss ways of obtaining a mode of grouped data. It is possible that more than one value may have the same maximum frequency. In such situations, the data is said to be multimodal. Though grouped data can also be multimodal, we shall restrict ourselves to problems having a single mode only.

Let us first recall how we found the mode for ungrouped data through the following example.

Example 4 : The wickets taken by a bowler in 10 cricket matches are as follows:

2 6 4 5 0 2 1 3 2 3

Find the mode of the data.

Solution : Let us form the frequency distribution table of the given data as follows:

Number of wickets	0	1	2	3	4	5	6
Number of matches	1	1	3	2	1	1	1

Clearly, 2 is the number of wickets taken by the bowler in the maximum number (i.e., 3) of matches. So, the mode of this data is 2.

In a grouped frequency distribution, it is not possible to determine the mode by looking at the frequencies. Here, we can only locate a class with the maximum frequency, called the **modal class**. The mode is a value inside the modal class, and is given by the formula:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

where l = lower limit of the modal class,

h = size of the class interval (assuming all class sizes to be equal),

f_1 = frequency of the modal class,

f_0 = frequency of the class preceding the modal class,

f_2 = frequency of the class succeeding the modal class.

Let us consider the following examples to illustrate the use of this formula.

Example 5 : A survey conducted on 20 households in a locality by a group of students resulted in the following frequency table for the number of family members in a household:

Family size	1 - 3	3 - 5	5 - 7	7 - 9	9 - 11
Number of families	7	8	2	2	1

Find the mode of this data.

Solution : Here the maximum class frequency is 8, and the class corresponding to this frequency is 3 – 5. So, the modal class is 3 – 5.

Now

modal class = 3 – 5, lower limit (l) of modal class = 3, class size (h) = 2

frequency (f_1) of the modal class = 8,

frequency (f_0) of class preceding the modal class = 7,

frequency (f_2) of class succeeding the modal class = 2.

Now, let us substitute these values in the formula :

$$\begin{aligned}\text{Mode} &= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h \\ &= 3 + \left(\frac{8 - 7}{2 \times 8 - 7 - 2} \right) \times 2 = 3 + \frac{2}{7} = 3.286\end{aligned}$$

Therefore, the mode of the data above is 3.286.

Example 6 : The marks distribution of 30 students in a mathematics examination are given in Table 13.3 of Example 1. Find the mode of this data. Also compare and interpret the mode and the mean.

Solution : Refer to Table 13.3 of Example 1. Since the maximum number of students (i.e., 7) have got marks in the interval 40 - 55, the modal class is 40 - 55. Therefore,

the lower limit (l) of the modal class = 40,

the class size (h) = 15,

the frequency (f_1) of modal class = 7,

the frequency (f_0) of the class preceding the modal class = 3,

the frequency (f_2) of the class succeeding the modal class = 6.

Now, using the formula:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h,$$

we get

$$\text{Mode} = 40 + \left(\frac{7 - 3}{14 - 6 - 3} \right) \times 15 = 52$$

So, the mode marks is 52.

Now, from Example 1, you know that the mean marks is 62.

So, the maximum number of students obtained 52 marks, while on an average a student obtained 62 marks.

Remarks :

1. In Example 6, the mode is less than the mean. But for some other problems it may be equal or more than the mean also.
2. It depends upon the demand of the situation whether we are interested in finding the average marks obtained by the students or the average of the marks obtained by most

of the students. In the first situation, the mean is required and in the second situation, the mode is required.

Activity 3 : Continuing with the same groups as formed in Activity 2 and the situations assigned to the groups. Ask each group to find the mode of the data. They should also compare this with the mean, and interpret the meaning of both.

Remark : The mode can also be calculated for grouped data with unequal class sizes. However, we shall not be discussing it.

EXERCISE 13.2

1. The following table shows the ages of the patients admitted in a hospital during a year:

Age (in years)	5 - 15	15 - 25	25 - 35	35 - 45	45 - 55	55 - 65
Number of patients	6	11	21	23	14	5

Find the mode and the mean of the data given above. Compare and interpret the two measures of central tendency.

2. The following data gives the information on the observed lifetimes (in hours) of 225 electrical components :

Lifetimes (in hours)	0 - 20	20 - 40	40 - 60	60 - 80	80 - 100	100 - 120
Frequency	10	35	52	61	38	29

Determine the modal lifetimes of the components.

3. The following data gives the distribution of total monthly household expenditure of 200 families of a village. Find the modal monthly expenditure of the families. Also, find the mean monthly expenditure :

Expenditure (in ₹)	Number of families
1000 - 1500	24
1500 - 2000	40
2000 - 2500	33
2500 - 3000	28
3000 - 3500	30
3500 - 4000	22
4000 - 4500	16
4500 - 5000	7

4. The following distribution gives the state-wise teacher-student ratio in higher secondary schools of India. Find the mode and mean of this data. Interpret the two measures.

Number of students per teacher	Number of states / U.T.
15 - 20	3
20 - 25	8
25 - 30	9
30 - 35	10
35 - 40	3
40 - 45	0
45 - 50	0
50 - 55	2

5. The given distribution shows the number of runs scored by some top batsmen of the world in one-day international cricket matches.

Runs scored	Number of batsmen
3000 - 4000	4
4000 - 5000	18
5000 - 6000	9
6000 - 7000	7
7000 - 8000	6
8000 - 9000	3
9000 - 10000	1
10000 - 11000	1

Find the mode of the data.

6. A student noted the number of cars passing through a spot on a road for 100 periods each of 3 minutes and summarised it in the table given below. Find the mode of the data :

Number of cars	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80
Frequency	7	14	13	12	20	11	15	8

13.4 Median of Grouped Data

As you have studied in Class IX, the median is a measure of central tendency which gives the value of the middle-most observation in the data. Recall that for finding the median of ungrouped data, we first arrange the data values of the observations in

ascending order. Then, if n is odd, the median is the $\left(\frac{n+1}{2}\right)$ th observation. And, if n

is even, then the median will be the average of the $\frac{n}{2}$ th and the $\left(\frac{n}{2} + 1\right)$ th observations.

Suppose, we have to find the median of the following data, which gives the marks, out of 50, obtained by 100 students in a test :

Marks obtained	20	29	28	33	42	38	43	25
Number of students	6	28	24	15	2	4	1	20

First, we arrange the marks in ascending order and prepare a frequency table as follows :

Table 13.9

Marks obtained	Number of students (Frequency)
20	6
25	20
28	24
29	28
33	15
38	4
42	2
43	1
Total	100

Here $n = 100$, which is even. The median will be the average of the $\frac{n}{2}$ th and the $\left(\frac{n}{2} + 1\right)$ th observations, i.e., the 50th and 51st observations. To find these observations, we proceed as follows:

Table 13.10

Marks obtained	Number of students
20	6
upto 25	$6 + 20 = 26$
upto 28	$26 + 24 = 50$
upto 29	$50 + 28 = 78$
upto 33	$78 + 15 = 93$
upto 38	$93 + 4 = 97$
upto 42	$97 + 2 = 99$
upto 43	$99 + 1 = 100$

Now we add another column depicting this information to the frequency table above and name it as *cumulative frequency column*.

Table 13.11

Marks obtained	Number of students	Cumulative frequency
20	6	6
25	20	26
28	24	50
29	28	78
33	15	93
38	4	97
42	2	99
43	1	100

From the table above, we see that:

50th observation is 28 (Why?)

51st observation is 29

So,
$$\text{Median} = \frac{28 + 29}{2} = 28.5$$

Remark : The part of Table 13.11 consisting Column 1 and Column 3 is known as *Cumulative Frequency Table*. The median marks 28.5 conveys the information that about 50% students obtained marks less than 28.5 and another 50% students obtained marks more than 28.5.

Now, let us see how to obtain the median of grouped data, through the following situation.

Consider a grouped frequency distribution of marks obtained, out of 100, by 53 students, in a certain examination, as follows:

Table 13.12

Marks	Number of students
0 - 10	5
10 - 20	3
20 - 30	4
30 - 40	3
40 - 50	3
50 - 60	4
60 - 70	7
70 - 80	9
80 - 90	7
90 - 100	8

From the table above, try to answer the following questions:

How many students have scored marks less than 10? The answer is clearly 5.

How many students have scored less than 20 marks? Observe that the number of students who have scored less than 20 include the number of students who have scored marks from 0 - 10 as well as the number of students who have scored marks from 10 - 20. So, the total number of students with marks less than 20 is $5 + 3$, i.e., 8. We say that the cumulative frequency of the class 10-20 is 8.

Similarly, we can compute the cumulative frequencies of the other classes, i.e., the number of students with marks less than 30, less than 40, . . . , less than 100. We give them in Table 13.13 given below:

Table 13.13

Marks obtained	Number of students (Cumulative frequency)
Less than 10	5
Less than 20	$5 + 3 = 8$
Less than 30	$8 + 4 = 12$
Less than 40	$12 + 3 = 15$
Less than 50	$15 + 3 = 18$
Less than 60	$18 + 4 = 22$
Less than 70	$22 + 7 = 29$
Less than 80	$29 + 9 = 38$
Less than 90	$38 + 7 = 45$
Less than 100	$45 + 8 = 53$

The distribution given above is called the *cumulative frequency distribution of the less than type*. Here 10, 20, 30, . . . 100, are the upper limits of the respective class intervals.

We can similarly make the table for the number of students with scores, more than or equal to 0, more than or equal to 10, more than or equal to 20, and so on. From Table 13.12, we observe that all 53 students have scored marks more than or equal to 0. Since there are 5 students scoring marks in the interval 0 - 10, this means that there are $53 - 5 = 48$ students getting more than or equal to 10 marks. Continuing in the same manner, we get the number of students scoring 20 or above as $48 - 3 = 45$, 30 or above as $45 - 4 = 41$, and so on, as shown in Table 13.14.

Table 13.14

Marks obtained	Number of students (Cumulative frequency)
More than or equal to 0	53
More than or equal to 10	$53 - 5 = 48$
More than or equal to 20	$48 - 3 = 45$
More than or equal to 30	$45 - 4 = 41$
More than or equal to 40	$41 - 3 = 38$
More than or equal to 50	$38 - 3 = 35$
More than or equal to 60	$35 - 4 = 31$
More than or equal to 70	$31 - 7 = 24$
More than or equal to 80	$24 - 9 = 15$
More than or equal to 90	$15 - 7 = 8$

The table above is called a *cumulative frequency distribution of the more than type*. Here 0, 10, 20, ..., 90 give the lower limits of the respective class intervals.

Now, to find the median of grouped data, we can make use of any of these cumulative frequency distributions.

Let us combine Tables 13.12 and 13.13 to get Table 13.15 given below:

Table 13.15

Marks	Number of students (f)	Cumulative frequency (cf)
0 - 10	5	5
10 - 20	3	8
20 - 30	4	12
30 - 40	3	15
40 - 50	3	18
50 - 60	4	22
60 - 70	7	29
70 - 80	9	38
80 - 90	7	45
90 - 100	8	53

Now in a grouped data, we may not be able to find the middle observation by looking at the cumulative frequencies as the middle observation will be some value in

a class interval. It is, therefore, necessary to find the value inside a class that divides the whole distribution into two halves. But which class should this be?

To find this class, we find the cumulative frequencies of all the classes and $\frac{n}{2}$.

We now locate the class whose cumulative frequency is greater than (and nearest to) $\frac{n}{2}$. This is called the *median class*. In the distribution above, $n = 53$. So, $\frac{n}{2} = 26.5$.

Now 60 – 70 is the class whose cumulative frequency 29 is greater than (and nearest to) $\frac{n}{2}$, i.e., 26.5.

Therefore, 60 – 70 is the **median class**.

After finding the median class, we use the following formula for calculating the median.

$$\text{Median} = l + \left(\frac{\frac{n}{2} - \text{cf}}{f} \right) \times h,$$

where

l = lower limit of median class,

n = number of observations,

cf = cumulative frequency of class preceding the median class,

f = frequency of median class,

h = class size (assuming class size to be equal).

Substituting the values $\frac{n}{2} = 26.5$, $l = 60$, $\text{cf} = 22$, $f = 7$, $h = 10$

in the formula above, we get

$$\begin{aligned}\text{Median} &= 60 + \left(\frac{26.5 - 22}{7} \right) \times 10 \\ &= 60 + \frac{45}{7} \\ &= 66.4\end{aligned}$$

So, about half the students have scored marks less than 66.4, and the other half have scored marks more than 66.4.

Example 7 : A survey regarding the heights (in cm) of 51 girls of Class X of a school was conducted and the following data was obtained:

Height (in cm)	Number of girls
Less than 140	4
Less than 145	11
Less than 150	29
Less than 155	40
Less than 160	46
Less than 165	51

Find the median height.

Solution : To calculate the median height, we need to find the class intervals and their corresponding frequencies.

The given distribution being of the *less than type*, 140, 145, 150, . . . , 165 give the upper limits of the corresponding class intervals. So, the classes should be below 140, 140 - 145, 145 - 150, . . . , 160 - 165. Observe that from the given distribution, we find that there are 4 girls with height less than 140, i.e., the frequency of class interval below 140 is 4. Now, there are 11 girls with heights less than 145 and 4 girls with height less than 140. Therefore, the number of girls with height in the interval 140 - 145 is $11 - 4 = 7$. Similarly, the frequency of 145 - 150 is $29 - 11 = 18$, for 150 - 155, it is $40 - 29 = 11$, and so on. So, our frequency distribution table with the given cumulative frequencies becomes:

Table 13.16

Class intervals	Frequency	Cumulative frequency
Below 140	4	4
140 - 145	7	11
145 - 150	18	29
150 - 155	11	40
155 - 160	6	46
160 - 165	5	51

Now $n = 51$. So, $\frac{n}{2} = \frac{51}{2} = 25.5$. This observation lies in the class 145 - 150. Then,

l (the lower limit) = 145,

cf (the cumulative frequency of the class preceding 145 - 150) = 11,

f (the frequency of the median class 145 - 150) = 18,

h (the class size) = 5.

Using the formula, Median = $l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$, we have

$$\begin{aligned} \text{Median} &= 145 + \left(\frac{25.5 - 11}{18} \right) \times 5 \\ &= 145 + \frac{72.5}{18} = 149.03. \end{aligned}$$

So, the median height of the girls is 149.03 cm.

This means that the height of about 50% of the girls is less than this height, and 50% are taller than this height.

Example 8 : The median of the following data is 525. Find the values of x and y , if the total frequency is 100.

Class intervals	Frequency
0 - 100	2
100 - 200	5
200 - 300	x
300 - 400	12
400 - 500	17
500 - 600	20
600 - 700	y
700 - 800	9
800 - 900	7
900 - 1000	4

Solution :

Class intervals	Frequency	Cumulative frequency
0 - 100	2	2
100 - 200	5	7
200 - 300	x	$7 + x$
300 - 400	12	$19 + x$
400 - 500	17	$36 + x$
500 - 600	20	$56 + x$
600 - 700	y	$56 + x + y$
700 - 800	9	$65 + x + y$
800 - 900	7	$72 + x + y$
900 - 1000	4	$76 + x + y$

It is given that $n = 100$

So, $76 + x + y = 100$, i.e., $x + y = 24$ (1)

The median is 525, which lies in the class 500 – 600

So, $l = 500$, $f = 20$, $cf = 36 + x$, $h = 100$

Using the formula : $\text{Median} = l + \left(\frac{\frac{n}{2} - cf}{f} \right) h$, we get

$$525 = 500 + \left(\frac{50 - 36 - x}{20} \right) \times 100$$

i.e., $525 - 500 = (14 - x) \times 5$

i.e., $25 = 70 - 5x$

i.e., $5x = 70 - 25 = 45$

So, $x = 9$

Therefore, from (1), we get $9 + y = 24$

i.e., $y = 15$

Now, that you have studied about all the three measures of central tendency, let us discuss **which measure would be best suited for a particular requirement.**

The mean is the most frequently used measure of central tendency because it takes into account all the observations, and lies between the extremes, i.e., the largest and the smallest observations of the entire data. It also enables us to compare two or more distributions. For example, by comparing the average (mean) results of students of different schools of a particular examination, we can conclude which school has a better performance.

However, extreme values in the data affect the mean. For example, the mean of classes having frequencies more or less the same is a good representative of the data. But, if one class has frequency, say 2, and the five others have frequency 20, 25, 20, 21, 18, then the mean will certainly not reflect the way the data behaves. So, in such cases, the mean is not a good representative of the data.

In problems where individual observations are not important, and we wish to find out a 'typical' observation, the median is more appropriate, e.g., finding the typical productivity rate of workers, average wage in a country, etc. These are situations where extreme values may be there. So, rather than the mean, we take the median as a better measure of central tendency.

In situations which require establishing the most frequent value or most popular item, the mode is the best choice, e.g., to find the most popular T.V. programme being watched, the consumer item in greatest demand, the colour of the vehicle used by most of the people, etc.

Remarks :

1. There is an empirical relationship between the three measures of central tendency :

$$3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$$

2. The median of grouped data with unequal class sizes can also be calculated. However, we shall not discuss it here.

EXERCISE 13.3

1. The following frequency distribution gives the monthly consumption of electricity of 68 consumers of a locality. Find the median, mean and mode of the data and compare them.

Monthly consumption (in units)	Number of consumers
65 - 85	4
85 - 105	5
105 - 125	13
125 - 145	20
145 - 165	14
165 - 185	8
185 - 205	4

2. If the median of the distribution given below is 28.5, find the values of x and y .

Class interval	Frequency
0 - 10	5
10 - 20	x
20 - 30	20
30 - 40	15
40 - 50	y
50 - 60	5
Total	60

3. A life insurance agent found the following data for distribution of ages of 100 policy holders. Calculate the median age, if policies are given only to persons having age 18 years onwards but less than 60 year.

Age (in years)	Number of policy holders
Below 20	2
Below 25	6
Below 30	24
Below 35	45
Below 40	78
Below 45	89
Below 50	92
Below 55	98
Below 60	100

4. The lengths of 40 leaves of a plant are measured correct to the nearest millimetre, and the data obtained is represented in the following table :

Length (in mm)	Number of leaves
118 - 126	3
127 - 135	5
136 - 144	9
145 - 153	12
154 - 162	5
163 - 171	4
172 - 180	2

Find the median length of the leaves.

(**Hint :** The data needs to be converted to continuous classes for finding the median, since the formula assumes continuous classes. The classes then change to 117.5 - 126.5, 126.5 - 135.5, ..., 171.5 - 180.5.)

5. The following table gives the distribution of the life time of 400 neon lamps :

Life time (in hours)	Number of lamps
1500 - 2000	14
2000 - 2500	56
2500 - 3000	60
3000 - 3500	86
3500 - 4000	74
4000 - 4500	62
4500 - 5000	48

Find the median life time of a lamp.

6. 100 surnames were randomly picked up from a local telephone directory and the frequency distribution of the number of letters in the English alphabets in the surnames was obtained as follows:

Number of letters	1 - 4	4 - 7	7 - 10	10 - 13	13 - 16	16 - 19
Number of surnames	6	30	40	16	4	4

Determine the median number of letters in the surnames. Find the mean number of letters in the surnames? Also, find the modal size of the surnames.

7. The distribution below gives the weights of 30 students of a class. Find the median weight of the students.

Weight (in kg)	40 - 45	45 - 50	50 - 55	55 - 60	60 - 65	65 - 70	70 - 75
Number of students	2	3	8	6	6	3	2

13.5 Summary

In this chapter, you have studied the following points:

1. The mean for grouped data can be found by :

(i) the direct method : $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$

(ii) the assumed mean method : $\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$

(iii) the step deviation method : $\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i} \right) \times h$,

with the assumption that the frequency of a class is centred at its mid-point, called its class mark.

2. The mode for grouped data can be found by using the formula:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

where symbols have their usual meanings.

3. The cumulative frequency of a class is the frequency obtained by adding the frequencies of all the classes preceding the given class.
4. The median for grouped data is formed by using the formula:

$$\text{Median} = l + \left(\frac{\frac{n}{2} - \text{cf}}{f} \right) \times h,$$

where symbols have their usual meanings.

A NOTE TO THE READER

For calculating mode and median for grouped data, it should be ensured that the class intervals are continuous before applying the formulae. Same condition also apply for construction of an ogive. Further, in case of ogives, the scale may not be the same on both the axes.



1062CH08

INTRODUCTION TO TRIGONOMETRY

8

There is perhaps nothing which so occupies the middle position of mathematics as trigonometry.

– J.F. Herbart (1890)

8.1 Introduction

You have already studied about triangles, and in particular, right triangles, in your earlier classes. Let us take some examples from our surroundings where right triangles can be imagined to be formed. For instance :

1. Suppose the students of a school are visiting Qutub Minar. Now, if a student is looking at the top of the Minar, a right triangle can be imagined to be made, as shown in Fig 8.1. Can the student find out the height of the Minar, without actually measuring it?
2. Suppose a girl is sitting on the balcony of her house located on the bank of a river. She is looking down at a flower pot placed on a stair of a temple situated nearby on the other bank of the river. A right triangle is imagined to be made in this situation as shown in Fig.8.2. If you know the height at which the person is sitting, can you find the width of the river?

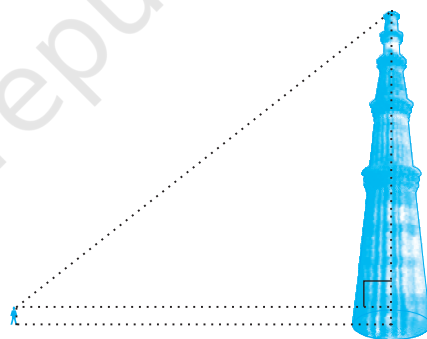


Fig. 8.1

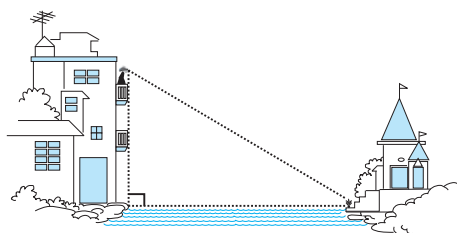


Fig. 8.2

3. Suppose a hot air balloon is flying in the air. A girl happens to spot the balloon in the sky and runs to her mother to tell her about it. Her mother rushes out of the house to look at the balloon. Now when the girl had spotted the balloon initially it was at point A. When both the mother and daughter came out to see it, it had already travelled to another point B. Can you find the altitude of B from the ground?

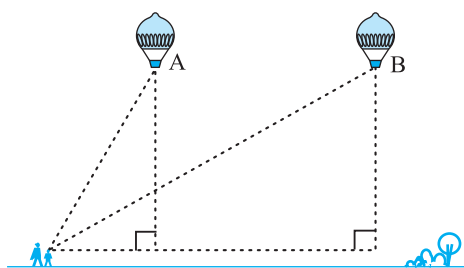


Fig. 8.3

In all the situations given above, the distances or heights can be found by using some mathematical techniques, which come under a branch of mathematics called ‘trigonometry’. The word ‘trigonometry’ is derived from the Greek words ‘tri’ (meaning three), ‘gon’ (meaning sides) and ‘metron’ (meaning measure). In fact, **trigonometry** is the study of relationships between the sides and angles of a triangle. The earliest known work on trigonometry was recorded in Egypt and Babylon. Early astronomers used it to find out the distances of the stars and planets from the Earth. Even today, most of the technologically advanced methods used in Engineering and Physical Sciences are based on trigonometrical concepts.

In this chapter, we will study some ratios of the sides of a right triangle with respect to its acute angles, called **trigonometric ratios of the angle**. We will restrict our discussion to acute angles only. However, these ratios can be extended to other angles also. We will also define the trigonometric ratios for angles of measure 0° and 90° . We will calculate trigonometric ratios for some specific angles and establish some identities involving these ratios, called **trigonometric identities**.

8.2 Trigonometric Ratios

In Section 8.1, you have seen some right triangles imagined to be formed in different situations.

Let us take a right triangle ABC as shown in Fig. 8.4.

Here, $\angle CAB$ (or, in brief, angle A) is an acute angle. Note the position of the side BC with respect to angle A. It faces $\angle A$. We call it the *side opposite* to angle A. AC is the *hypotenuse* of the right triangle and the side AB is a part of $\angle A$. So, we call it the *side adjacent* to angle A.

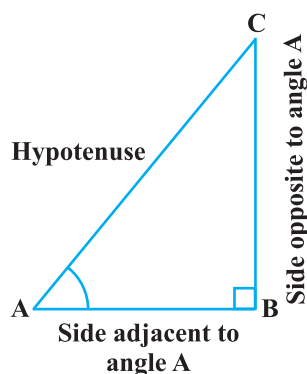


Fig. 8.4

Note that the position of sides change when you consider angle C in place of A (see Fig. 8.5).

You have studied the concept of ‘ratio’ in your earlier classes. We now define certain ratios involving the sides of a right triangle, and call them trigonometric ratios.

The trigonometric ratios of the angle A in right triangle ABC (see Fig. 8.4) are defined as follows :

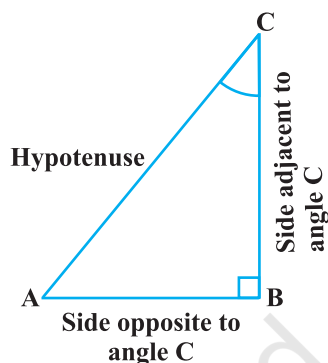


Fig. 8.5

$$\text{sine of } \angle A = \frac{\text{side opposite to angle A}}{\text{hypotenuse}} = \frac{BC}{AC}$$

$$\text{cosine of } \angle A = \frac{\text{side adjacent to angle A}}{\text{hypotenuse}} = \frac{AB}{AC}$$

$$\text{tangent of } \angle A = \frac{\text{side opposite to angle A}}{\text{side adjacent to angle A}} = \frac{BC}{AB}$$

$$\text{cosecant of } \angle A = \frac{1}{\text{sine of } \angle A} = \frac{\text{hypotenuse}}{\text{side opposite to angle A}} = \frac{AC}{BC}$$

$$\text{secant of } \angle A = \frac{1}{\text{cosine of } \angle A} = \frac{\text{hypotenuse}}{\text{side adjacent to angle A}} = \frac{AC}{AB}$$

$$\text{cotangent of } \angle A = \frac{1}{\text{tangent of } \angle A} = \frac{\text{side adjacent to angle A}}{\text{side opposite to angle A}} = \frac{AB}{BC}$$

The ratios defined above are abbreviated as $\sin A$, $\cos A$, $\tan A$, $\text{cosec } A$, $\sec A$ and $\cot A$ respectively. Note that the ratios **cosec A**, **sec A** and **cot A** are respectively, the reciprocals of the ratios $\sin A$, $\cos A$ and $\tan A$.

$$\text{Also, observe that } \tan A = \frac{BC}{AB} = \frac{\frac{BC}{AC}}{\frac{AB}{AC}} = \frac{\sin A}{\cos A} \text{ and } \cot A = \frac{\cos A}{\sin A}.$$

So, the **trigonometric ratios** of an acute angle in a right triangle express the relationship between the angle and the length of its sides.

Why don't you try to define the trigonometric ratios for angle C in the right triangle? (See Fig. 8.5)

The first use of the idea of ‘sine’ in the way we use it today was in the work *Aryabhatiyam* by Aryabhata, in A.D. 500. Aryabhata used the word **ardha-jya** for the half-chord, which was shortened to **jya** or **jiva** in due course. When the *Aryabhatiyam* was translated into Arabic, the word **jiva** was retained as it is. The word **jiva** was translated into **sinus**, which means curve, when the Arabic version was translated into Latin. Soon the word **sinus**, also used as **sine**, became common in mathematical texts throughout Europe. An English Professor of astronomy Edmund Gunter (1581–1626), first used the abbreviated notation ‘**sin**’.



Aryabhata
C.E. 476 – 550

The origin of the terms ‘**cosine**’ and ‘**tangent**’ was much later. The cosine function arose from the need to compute the sine of the complementary angle. Aryabhata called it **kotijya**. The name **cosinus** originated with Edmund Gunter. In 1674, the English Mathematician Sir Jonas Moore first used the abbreviated notation ‘**cos**’.

Remark : Note that the symbol $\sin A$ is used as an abbreviation for ‘the sine of the angle A ’. $\sin A$ is *not* the product of ‘sin’ and A . ‘sin’ separated from A has no meaning. Similarly, $\cos A$ is *not* the product of ‘cos’ and A . Similar interpretations follow for other trigonometric ratios also.

Now, if we take a point P on the hypotenuse AC or a point Q on AC extended, of the right triangle ABC and draw PM perpendicular to AB and QN perpendicular to AB extended (see Fig. 8.6), how will the trigonometric ratios of $\angle A$ in $\triangle PAM$ differ from those of $\angle A$ in $\triangle CAB$ or from those of $\angle A$ in $\triangle QAN$?

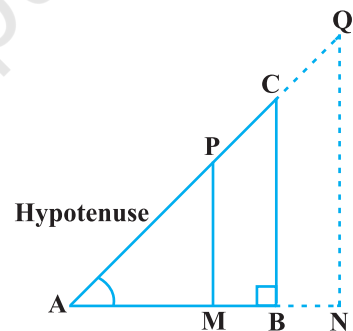


Fig. 8.6

To answer this, first look at these triangles. Is $\triangle PAM$ similar to $\triangle CAB$? From Chapter 6, recall the AA similarity criterion. Using the criterion, you will see that the triangles PAM and CAB are similar. Therefore, by the property of similar triangles, the corresponding sides of the triangles are proportional.

So, we have

$$\frac{AM}{AB} = \frac{AP}{AC} = \frac{MP}{BC}.$$

From this, we find $\frac{MP}{AP} = \frac{BC}{AC} = \sin A$.

Similarly, $\frac{AM}{AP} = \frac{AB}{AC} = \cos A$, $\frac{MP}{AM} = \frac{BC}{AB} = \tan A$ and so on.

This shows that the trigonometric ratios of angle A in $\triangle PAM$ not differ from those of angle A in $\triangle CAB$.

In the same way, you should check that the value of $\sin A$ (and also of other trigonometric ratios) remains the same in $\triangle QAN$ also.

From our observations, it is now clear that **the values of the trigonometric ratios of an angle do not vary with the lengths of the sides of the triangle, if the angle remains the same.**

Note : For the sake of convenience, we may write $\sin^2 A$, $\cos^2 A$, etc., in place of $(\sin A)^2$, $(\cos A)^2$, etc., respectively. But $\operatorname{cosec} A = (\sin A)^{-1} \neq \sin^{-1} A$ (it is called sine inverse A). $\sin^{-1} A$ has a different meaning, which will be discussed in higher classes. Similar conventions hold for the other trigonometric ratios as well. Sometimes, the Greek letter θ (theta) is also used to denote an angle.

We have defined six trigonometric ratios of an acute angle. If we know any one of the ratios, can we obtain the other ratios? Let us see.

If in a right triangle ABC , $\sin A = \frac{1}{3}$,
then this means that $\frac{BC}{AC} = \frac{1}{3}$, i.e., the
lengths of the sides BC and AC of the triangle
 ABC are in the ratio $1 : 3$ (see Fig. 8.7). So if
 BC is equal to k , then AC will be $3k$, where
 k is any positive number. To determine other
trigonometric ratios for the angle A , we need to find the length of the third side
 AB . Do you remember the Pythagoras theorem? Let us use it to determine the
required length AB .

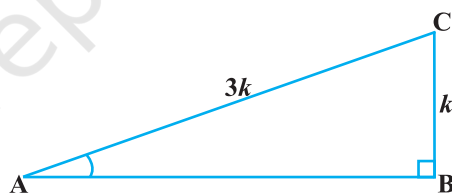


Fig. 8.7

$$AB^2 = AC^2 - BC^2 = (3k)^2 - (k)^2 = 8k^2 = (2\sqrt{2} k)^2$$

Therefore, $AB = \pm 2\sqrt{2} k$

So, we get $AB = 2\sqrt{2} k$ (Why is AB not $-2\sqrt{2} k$?)

Now, $\cos A = \frac{AB}{AC} = \frac{2\sqrt{2} k}{3k} = \frac{2\sqrt{2}}{3}$

Similarly, you can obtain the other trigonometric ratios of the angle A .

Remark : Since the hypotenuse is the longest side in a right triangle, the value of $\sin A$ or $\cos A$ is always less than 1 (or, in particular, equal to 1).

Let us consider some examples.

Example 1 : Given $\tan A = \frac{4}{3}$, find the other trigonometric ratios of the angle A .

Solution : Let us first draw a right $\triangle ABC$ (see Fig 8.8).

Now, we know that $\tan A = \frac{BC}{AB} = \frac{4}{3}$.

Therefore, if $BC = 4k$, then $AB = 3k$, where k is a positive number.

Now, by using the Pythagoras Theorem, we have

$$AC^2 = AB^2 + BC^2 = (4k)^2 + (3k)^2 = 25k^2$$

So, $AC = 5k$

Now, we can write all the trigonometric ratios using their definitions.

$$\sin A = \frac{BC}{AC} = \frac{4k}{5k} = \frac{4}{5}$$

$$\cos A = \frac{AB}{AC} = \frac{3k}{5k} = \frac{3}{5}$$

Therefore, $\cot A = \frac{1}{\tan A} = \frac{3}{4}$, $\operatorname{cosec} A = \frac{1}{\sin A} = \frac{5}{4}$ and $\sec A = \frac{1}{\cos A} = \frac{5}{3}$.

Example 2 : If $\angle B$ and $\angle Q$ are acute angles such that $\sin B = \sin Q$, then prove that $\angle B = \angle Q$.

Solution : Let us consider two right triangles ABC and PQR where $\sin B = \sin Q$ (see Fig. 8.9).

We have

$$\sin B = \frac{AC}{AB}$$

and

$$\sin Q = \frac{PR}{PQ}$$

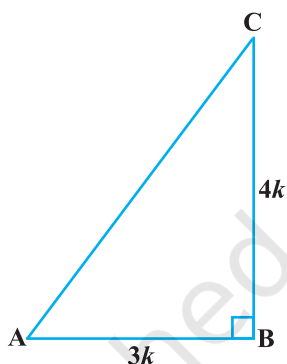


Fig. 8.8

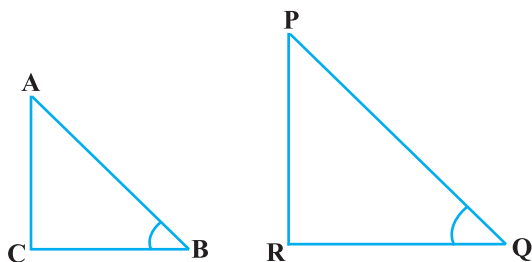


Fig. 8.9

Then
$$\frac{AC}{AB} = \frac{PR}{PQ}$$

Therefore,
$$\frac{AC}{PR} = \frac{AB}{PQ} = k, \text{ say} \quad (1)$$

Now, using Pythagoras theorem,

$$BC = \sqrt{AB^2 - AC^2}$$

and

$$QR = \sqrt{PQ^2 - PR^2}$$

So,
$$\frac{BC}{QR} = \frac{\sqrt{AB^2 - AC^2}}{\sqrt{PQ^2 - PR^2}} = \frac{\sqrt{k^2 PQ^2 - k^2 PR^2}}{\sqrt{PQ^2 - PR^2}} = \frac{k \sqrt{PQ^2 - PR^2}}{\sqrt{PQ^2 - PR^2}} = k \quad (2)$$

From (1) and (2), we have

$$\frac{AC}{PR} = \frac{AB}{PQ} = \frac{BC}{QR}$$

Then, by using Theorem 6.4, $\Delta ACB \sim \Delta PRQ$ and therefore, $\angle B = \angle Q$.

Example 3 : Consider ΔACB , right-angled at C, in which $AB = 29$ units, $BC = 21$ units and $\angle ABC = \theta$ (see Fig. 8.10). Determine the values of

(i) $\cos^2 \theta + \sin^2 \theta$,

(ii) $\cos^2 \theta - \sin^2 \theta$.

Solution : In ΔACB , we have

$$\begin{aligned} AC &= \sqrt{AB^2 - BC^2} = \sqrt{(29)^2 - (21)^2} \\ &= \sqrt{(29 - 21)(29 + 21)} = \sqrt{(8)(50)} = \sqrt{400} = 20 \text{ units} \end{aligned}$$

So,
$$\sin \theta = \frac{AC}{AB} = \frac{20}{29}, \cos \theta = \frac{BC}{AB} = \frac{21}{29}.$$

Now, (i) $\cos^2 \theta + \sin^2 \theta = \left(\frac{20}{29}\right)^2 + \left(\frac{21}{29}\right)^2 = \frac{20^2 + 21^2}{29^2} = \frac{400 + 441}{841} = 1,$

and (ii) $\cos^2 \theta - \sin^2 \theta = \left(\frac{21}{29}\right)^2 - \left(\frac{20}{29}\right)^2 = \frac{(21 + 20)(21 - 20)}{29^2} = \frac{41}{841}.$

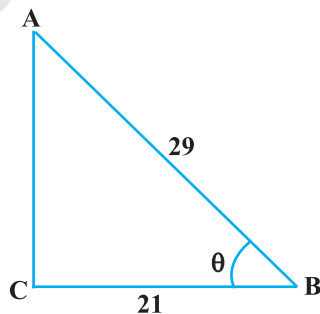


Fig. 8.10

Example 4 : In a right triangle ABC, right-angled at B, if $\tan A = 1$, then verify that

$$2 \sin A \cos A = 1.$$

Solution : In $\triangle ABC$, $\tan A = \frac{BC}{AB} = 1$ (see Fig 8.11)

i.e., $BC = AB$

Let $AB = BC = k$, where k is a positive number.

$$\begin{aligned} \text{Now, } AC &= \sqrt{AB^2 + BC^2} \\ &= \sqrt{(k)^2 + (k)^2} = k\sqrt{2} \end{aligned}$$

Therefore, $\sin A = \frac{BC}{AC} = \frac{1}{\sqrt{2}}$ and $\cos A = \frac{AB}{AC} = \frac{1}{\sqrt{2}}$

So, $2 \sin A \cos A = 2 \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) = 1$, which is the required value.

Example 5 : In $\triangle OPQ$, right-angled at P, $OP = 7$ cm and $OQ - PQ = 1$ cm (see Fig. 8.12). Determine the values of $\sin Q$ and $\cos Q$.

Solution : In $\triangle OPQ$, we have

$$OQ^2 = OP^2 + PQ^2$$

i.e., $(1 + PQ)^2 = OP^2 + PQ^2$ (Why?)

i.e., $1 + PQ^2 + 2PQ = OP^2 + PQ^2$

i.e., $1 + 2PQ = 7^2$ (Why?)

i.e., $PQ = 24$ cm and $OQ = 1 + PQ = 25$ cm

So, $\sin Q = \frac{7}{25}$ and $\cos Q = \frac{24}{25}$.

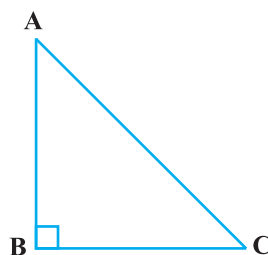


Fig. 8.11

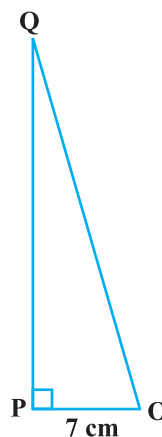


Fig. 8.12

EXERCISE 8.1

- In $\triangle ABC$, right-angled at B, $AB = 24$ cm, $BC = 7$ cm. Determine :
 - $\sin A$, $\cos A$
 - $\sin C$, $\cos C$
- In Fig. 8.13, find $\tan P - \cot R$.
- If $\sin A = \frac{3}{4}$, calculate $\cos A$ and $\tan A$.
- Given $15 \cot A = 8$, find $\sin A$ and $\sec A$.
- Given $\sec \theta = \frac{13}{12}$, calculate all other trigonometric ratios.
- If $\angle A$ and $\angle B$ are acute angles such that $\cos A = \cos B$, then show that $\angle A = \angle B$.
- If $\cot \theta = \frac{7}{8}$, evaluate : (i) $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$, (ii) $\cot^2 \theta$
- If $3 \cot A = 4$, check whether $\frac{1 - \tan^2 A}{1 + \tan^2 A} = \cos^2 A - \sin^2 A$ or not.
- In triangle ABC, right-angled at B, if $\tan A = \frac{1}{\sqrt{3}}$, find the value of:
 - $\sin A \cos C + \cos A \sin C$
 - $\cos A \cos C - \sin A \sin C$
- In $\triangle PQR$, right-angled at Q, $PR + QR = 25$ cm and $PQ = 5$ cm. Determine the values of $\sin P$, $\cos P$ and $\tan P$.
- State whether the following are true or false. Justify your answer.
 - The value of $\tan A$ is always less than 1.
 - $\sec A = \frac{12}{5}$ for some value of angle A.
 - $\cos A$ is the abbreviation used for the cosecant of angle A.
 - $\cot A$ is the product of $\cot$ and A.
 - $\sin \theta = \frac{4}{3}$ for some angle θ .

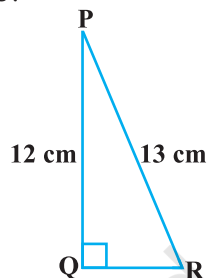


Fig. 8.13

8.3 Trigonometric Ratios of Some Specific Angles

From geometry, you are already familiar with the construction of angles of 30° , 45° , 60° and 90° . In this section, we will find the values of the trigonometric ratios for these angles and, of course, for 0° .

Trigonometric Ratios of 45°

In $\triangle ABC$, right-angled at B, if one angle is 45° , then the other angle is also 45° , i.e., $\angle A = \angle C = 45^\circ$ (see Fig. 8.14).

So, $BC = AB$ (Why?)

Now, Suppose $BC = AB = a$.

Then by Pythagoras Theorem, $AC^2 = AB^2 + BC^2 = a^2 + a^2 = 2a^2$,

and, therefore, $AC = a\sqrt{2}$.

Using the definitions of the trigonometric ratios, we have :

$$\sin 45^\circ = \frac{\text{side opposite to angle } 45^\circ}{\text{hypotenuse}} = \frac{BC}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{\text{side adjacent to angle } 45^\circ}{\text{hypotenuse}} = \frac{AB}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = \frac{\text{side opposite to angle } 45^\circ}{\text{side adjacent to angle } 45^\circ} = \frac{BC}{AB} = \frac{a}{a} = 1$$

$$\text{Also, } \operatorname{cosec} 45^\circ = \frac{1}{\sin 45^\circ} = \sqrt{2}, \sec 45^\circ = \frac{1}{\cos 45^\circ} = \sqrt{2}, \cot 45^\circ = \frac{1}{\tan 45^\circ} = 1.$$

Trigonometric Ratios of 30° and 60°

Let us now calculate the trigonometric ratios of 30° and 60° . Consider an equilateral triangle ABC. Since each angle in an equilateral triangle is 60° , therefore, $\angle A = \angle B = \angle C = 60^\circ$.

Draw the perpendicular AD from A to the side BC (see Fig. 8.15).

Now $\triangle ABD \cong \triangle ACD$ (Why?)

Therefore, $BD = DC$

and $\angle BAD = \angle CAD$ (CPCT)

Now observe that:

$\triangle ABD$ is a right triangle, right-angled at D with $\angle BAD = 30^\circ$ and $\angle ABD = 60^\circ$ (see Fig. 8.15).

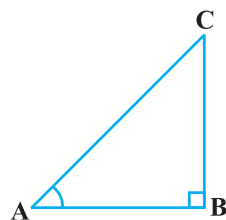


Fig. 8.14

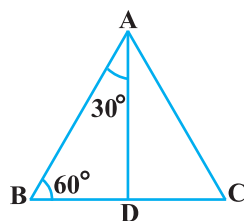


Fig. 8.15

As you know, for finding the trigonometric ratios, we need to know the lengths of the sides of the triangle. So, let us suppose that $AB = 2a$.

Then,
$$BD = \frac{1}{2}BC = a$$

and
$$AD^2 = AB^2 - BD^2 = (2a)^2 - (a)^2 = 3a^2,$$

Therefore,
$$AD = a\sqrt{3}$$

Now, we have :

$$\sin 30^\circ = \frac{BD}{AB} = \frac{a}{2a} = \frac{1}{2}, \quad \cos 30^\circ = \frac{AD}{AB} = \frac{a\sqrt{3}}{2a} = \frac{\sqrt{3}}{2}$$

$$\tan 30^\circ = \frac{BD}{AD} = \frac{a}{a\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

Also,
$$\operatorname{cosec} 30^\circ = \frac{1}{\sin 30^\circ} = 2, \quad \sec 30^\circ = \frac{1}{\cos 30^\circ} = \frac{2}{\sqrt{3}}$$

$$\cot 30^\circ = \frac{1}{\tan 30^\circ} = \sqrt{3}.$$

Similarly,

$$\sin 60^\circ = \frac{AD}{AB} = \frac{a\sqrt{3}}{2a} = \frac{\sqrt{3}}{2}, \quad \cos 60^\circ = \frac{1}{2}, \quad \tan 60^\circ = \sqrt{3},$$

$$\operatorname{cosec} 60^\circ = \frac{2}{\sqrt{3}}, \quad \sec 60^\circ = 2 \text{ and } \cot 60^\circ = \frac{1}{\sqrt{3}}.$$

Trigonometric Ratios of 0° and 90°

Let us see what happens to the trigonometric ratios of angle A , if it is made smaller and smaller in the right triangle ABC (see Fig. 8.16), till it becomes zero. As $\angle A$ gets smaller and smaller, the length of the side BC decreases. The point C gets closer to point B , and finally when $\angle A$ becomes very close to 0° , AC becomes almost the same as AB (see Fig. 8.17).

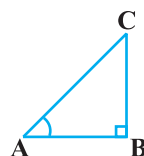


Fig. 8.16

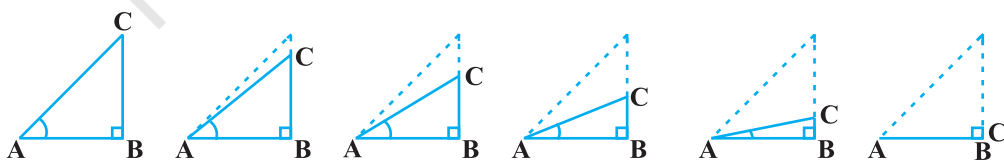


Fig. 8.17

When $\angle A$ is very close to 0° , BC gets very close to 0 and so the value of $\sin A = \frac{BC}{AC}$ is very close to 0. Also, when $\angle A$ is very close to 0° , AC is nearly the same as AB and so the value of $\cos A = \frac{AB}{AC}$ is very close to 1.

This helps us to see how we can define the values of $\sin A$ and $\cos A$ when $A = 0^\circ$. We define : **$\sin 0^\circ = 0$ and $\cos 0^\circ = 1$.**

Using these, we have :

$$\tan 0^\circ = \frac{\sin 0^\circ}{\cos 0^\circ} = 0, \cot 0^\circ = \frac{1}{\tan 0^\circ}, \text{ which is not defined. (Why?)}$$

$$\sec 0^\circ = \frac{1}{\cos 0^\circ} = 1 \text{ and } \operatorname{cosec} 0^\circ = \frac{1}{\sin 0^\circ}, \text{ which is again not defined. (Why?)}$$

Now, let us see what happens to the trigonometric ratios of $\angle A$, when it is made larger and larger in $\triangle ABC$ till it becomes 90° . As $\angle A$ gets larger and larger, $\angle C$ gets smaller and smaller. Therefore, as in the case above, the length of the side AB goes on decreasing. The point A gets closer to point B . Finally when $\angle A$ is very close to 90° , $\angle C$ becomes very close to 0° and the side AC almost coincides with side BC (see Fig. 8.18).

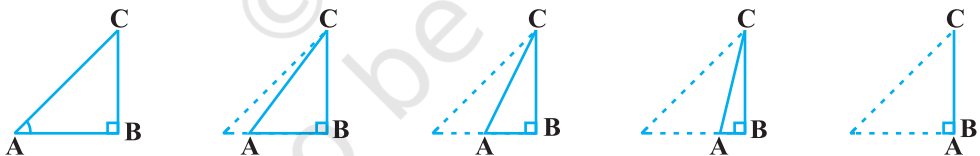


Fig. 8.18

When $\angle C$ is very close to 0° , $\angle A$ is very close to 90° , side AC is nearly the same as side BC , and so $\sin A$ is very close to 1. Also when $\angle A$ is very close to 90° , $\angle C$ is very close to 0° , and the side AB is nearly zero, so $\cos A$ is very close to 0.

So, we define : **$\sin 90^\circ = 1$ and $\cos 90^\circ = 0$.**

Now, why don't you find the other trigonometric ratios of 90° ?

We shall now give the values of all the trigonometric ratios of 0° , 30° , 45° , 60° and 90° in Table 8.1, for ready reference.

Table 8.1

$\angle A$	0°	30°	45°	60°	90°
$\sin A$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos A$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan A$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$\operatorname{cosec} A$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec A$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\cot A$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

Remark : From the table above you can observe that as $\angle A$ increases from 0° to 90° , $\sin A$ increases from 0 to 1 and $\cos A$ decreases from 1 to 0.

Let us illustrate the use of the values in the table above through some examples.

Example 6 : In $\triangle ABC$, right-angled at B, $AB = 5$ cm and $\angle ACB = 30^\circ$ (see Fig. 8.19). Determine the lengths of the sides BC and AC.

Solution : To find the length of the side BC, we will choose the trigonometric ratio involving BC and the given side AB. Since BC is the side adjacent to angle C and AB is the side opposite to angle C, therefore

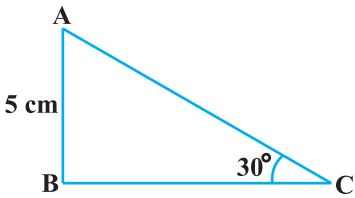


Fig. 8.19

$$\frac{AB}{BC} = \tan C$$

i.e.,
$$\frac{5}{BC} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

which gives
$$BC = 5\sqrt{3} \text{ cm}$$

To find the length of the side AC, we consider

$$\sin 30^\circ = \frac{AB}{AC} \quad (\text{Why?})$$

$$\text{i.e.,} \quad \frac{1}{2} = \frac{5}{AC}$$

$$\text{i.e.,} \quad AC = 10 \text{ cm}$$

Note that alternatively we could have used Pythagoras theorem to determine the third side in the example above,

$$\text{i.e.,} \quad AC = \sqrt{AB^2 + BC^2} = \sqrt{5^2 + (5\sqrt{3})^2} \text{ cm} = 10 \text{ cm}.$$

Example 7 : In ΔPQR , right-angled at Q (see Fig. 8.20), $PQ = 3 \text{ cm}$ and $PR = 6 \text{ cm}$. Determine $\angle QPR$ and $\angle PRQ$.

Solution : Given $PQ = 3 \text{ cm}$ and $PR = 6 \text{ cm}$.

Therefore,

$$\frac{PQ}{PR} = \sin R$$

$$\text{or} \quad \sin R = \frac{3}{6} = \frac{1}{2}$$

$$\text{So,} \quad \angle PRQ = 30^\circ$$

$$\text{and therefore,} \quad \angle QPR = 60^\circ. \quad (\text{Why?})$$

You may note that if one of the sides and any other part (either an acute angle or any side) of a right triangle is known, the remaining sides and angles of the triangle can be determined.

Example 8 : If $\sin (A - B) = \frac{1}{2}$, $\cos (A + B) = \frac{1}{2}$, $0^\circ < A + B \leq 90^\circ$, $A > B$, find A and B.

$$\text{Solution :} \quad \text{Since, } \sin (A - B) = \frac{1}{2}, \text{ therefore, } A - B = 30^\circ \quad (\text{Why?}) \quad (1)$$

$$\text{Also, since } \cos (A + B) = \frac{1}{2}, \text{ therefore, } A + B = 60^\circ \quad (\text{Why?}) \quad (2)$$

Solving (1) and (2), we get : $A = 45^\circ$ and $B = 15^\circ$.

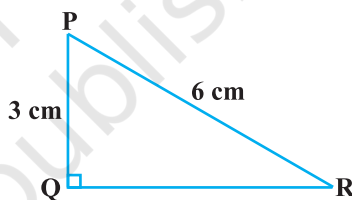


Fig. 8.20

EXERCISE 8.2

1. Evaluate the following :

(i) $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$

(ii) $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$

(iii) $\frac{\cos 45^\circ}{\sec 30^\circ + \operatorname{cosec} 30^\circ}$

(iv) $\frac{\sin 30^\circ + \tan 45^\circ - \operatorname{cosec} 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$

(v) $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$

2. Choose the correct option and justify your choice :

(i) $\frac{2 \tan 30^\circ}{1 + \tan^2 30^\circ} =$

(A) $\sin 60^\circ$

(B) $\cos 60^\circ$

(C) $\tan 60^\circ$

(D) $\sin 30^\circ$

(ii) $\frac{1 - \tan^2 45^\circ}{1 + \tan^2 45^\circ} =$

(A) $\tan 90^\circ$

(B) 1

(C) $\sin 45^\circ$

(D) 0

(iii) $\sin 2A = 2 \sin A$ is true when $A =$

(A) 0°

(B) 30°

(C) 45°

(D) 60°

(iv) $\frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ} =$

(A) $\cos 60^\circ$

(B) $\sin 60^\circ$

(C) $\tan 60^\circ$

(D) $\sin 30^\circ$

3. If $\tan (A + B) = \sqrt{3}$ and $\tan (A - B) = \frac{1}{\sqrt{3}}$; $0^\circ < A + B \leq 90^\circ$; $A > B$, find A and B.

4. State whether the following are true or false. Justify your answer.

(i) $\sin (A + B) = \sin A + \sin B$.

(ii) The value of $\sin \theta$ increases as θ increases.

(iii) The value of $\cos \theta$ increases as θ increases.

(iv) $\sin \theta = \cos \theta$ for all values of θ .

(v) $\cot A$ is not defined for $A = 0^\circ$.

8.4 Trigonometric Identities

You may recall that an equation is called an identity when it is true for all values of the variables involved. Similarly, an equation involving trigonometric ratios of an angle is called a **trigonometric identity**, if it is true for all values of the angle(s) involved.

In this section, we will prove one trigonometric identity, and use it further to prove other useful trigonometric identities.

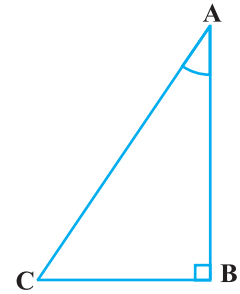


Fig. 8.21

In $\triangle ABC$, right-angled at B (see Fig. 8.21), we have:

$$AB^2 + BC^2 = AC^2 \quad (1)$$

Dividing each term of (1) by AC^2 , we get

$$\frac{AB^2}{AC^2} + \frac{BC^2}{AC^2} = \frac{AC^2}{AC^2}$$

$$\text{i.e.,} \quad \left(\frac{AB}{AC}\right)^2 + \left(\frac{BC}{AC}\right)^2 = \left(\frac{AC}{AC}\right)^2$$

$$\text{i.e.,} \quad (\cos A)^2 + (\sin A)^2 = 1$$

$$\text{i.e.,} \quad \cos^2 A + \sin^2 A = 1 \quad (2)$$

This is true for all A such that $0^\circ \leq A \leq 90^\circ$. So, this is a trigonometric identity.

Let us now divide (1) by AB^2 . We get

$$\frac{AB^2}{AB^2} + \frac{BC^2}{AB^2} = \frac{AC^2}{AB^2}$$

$$\text{or,} \quad \left(\frac{AB}{AB}\right)^2 + \left(\frac{BC}{AB}\right)^2 = \left(\frac{AC}{AB}\right)^2$$

$$\text{i.e.,} \quad 1 + \tan^2 A = \sec^2 A \quad (3)$$

Is this equation true for $A = 0^\circ$? Yes, it is. What about $A = 90^\circ$? Well, $\tan A$ and $\sec A$ are not defined for $A = 90^\circ$. So, (3) is true for all A such that $0^\circ \leq A < 90^\circ$.

Let us see what we get on dividing (1) by BC^2 . We get

$$\frac{AB^2}{BC^2} + \frac{BC^2}{BC^2} = \frac{AC^2}{BC^2}$$

$$\text{i.e.,} \quad \left(\frac{AB}{BC}\right)^2 + \left(\frac{BC}{BC}\right)^2 = \left(\frac{AC}{BC}\right)^2$$

$$\text{i.e.,} \quad \cot^2 A + 1 = \operatorname{cosec}^2 A \quad (4)$$

Note that cosec A and cot A are not defined for $A = 0^\circ$. Therefore (4) is true for all A such that $0^\circ < A \leq 90^\circ$.

Using these identities, we can express each trigonometric ratio in terms of other trigonometric ratios, i.e., if any one of the ratios is known, we can also determine the values of other trigonometric ratios.

Let us see how we can do this using these identities. Suppose we know that

$$\tan A = \frac{1}{\sqrt{3}}. \text{ Then, } \cot A = \sqrt{3}.$$

$$\text{Since, } \sec^2 A = 1 + \tan^2 A = 1 + \frac{1}{3} = \frac{4}{3}, \sec A = \frac{2}{\sqrt{3}}, \text{ and } \cos A = \frac{\sqrt{3}}{2}.$$

$$\text{Again, } \sin A = \sqrt{1 - \cos^2 A} = \sqrt{1 - \frac{3}{4}} = \frac{1}{2}. \text{ Therefore, } \operatorname{cosec} A = 2.$$

Example 9 : Express the ratios $\cos A$, $\tan A$ and $\sec A$ in terms of $\sin A$.

Solution : Since $\cos^2 A + \sin^2 A = 1$, therefore,

$$\cos^2 A = 1 - \sin^2 A, \text{ i.e., } \cos A = \pm \sqrt{1 - \sin^2 A}$$

$$\text{This gives} \quad \cos A = \sqrt{1 - \sin^2 A} \quad (\text{Why?})$$

$$\text{Hence,} \quad \tan A = \frac{\sin A}{\cos A} = \frac{\sin A}{\sqrt{1 - \sin^2 A}} \text{ and } \sec A = \frac{1}{\cos A} = \frac{1}{\sqrt{1 - \sin^2 A}}$$

Example 10 : Prove that $\sec A (1 - \sin A)(\sec A + \tan A) = 1$.

Solution :

$$\text{LHS} = \sec A (1 - \sin A)(\sec A + \tan A) = \left(\frac{1}{\cos A}\right)(1 - \sin A)\left(\frac{1}{\cos A} + \frac{\sin A}{\cos A}\right)$$

$$\begin{aligned}
 &= \frac{(1 - \sin A)(1 + \sin A)}{\cos^2 A} = \frac{1 - \sin^2 A}{\cos^2 A} \\
 &= \frac{\cos^2 A}{\cos^2 A} = 1 = \text{RHS}
 \end{aligned}$$

Example 11 : Prove that $\frac{\cot A - \cos A}{\cot A + \cos A} = \frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1}$

Solution : LHS = $\frac{\cot A - \cos A}{\cot A + \cos A} = \frac{\frac{\cos A}{\sin A} - \cos A}{\frac{\cos A}{\sin A} + \cos A}$

$$\begin{aligned}
 &= \frac{\cos A \left(\frac{1}{\sin A} - 1 \right)}{\cos A \left(\frac{1}{\sin A} + 1 \right)} = \frac{\left(\frac{1}{\sin A} - 1 \right)}{\left(\frac{1}{\sin A} + 1 \right)} = \frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1} = \text{RHS}
 \end{aligned}$$

Example 12 : Prove that $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{1}{\sec \theta - \tan \theta}$, using the identity $\sec^2 \theta = 1 + \tan^2 \theta$.

Solution : Since we will apply the identity involving $\sec \theta$ and $\tan \theta$, let us first convert the LHS (of the identity we need to prove) in terms of $\sec \theta$ and $\tan \theta$ by dividing numerator and denominator by $\cos \theta$.

$$\begin{aligned}
 \text{LHS} &= \frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{\tan \theta - 1 + \sec \theta}{\tan \theta + 1 - \sec \theta} \\
 &= \frac{(\tan \theta + \sec \theta) - 1}{(\tan \theta - \sec \theta) + 1} = \frac{\{(\tan \theta + \sec \theta) - 1\} (\tan \theta - \sec \theta)}{\{(\tan \theta - \sec \theta) + 1\} (\tan \theta - \sec \theta)} \\
 &= \frac{(\tan^2 \theta - \sec^2 \theta) - (\tan \theta - \sec \theta)}{\{\tan \theta - \sec \theta + 1\} (\tan \theta - \sec \theta)} \\
 &= \frac{-1 - \tan \theta + \sec \theta}{(\tan \theta - \sec \theta + 1) (\tan \theta - \sec \theta)}
 \end{aligned}$$

$$= \frac{-1}{\tan \theta - \sec \theta} = \frac{1}{\sec \theta - \tan \theta},$$

which is the RHS of the identity, we are required to prove.

EXERCISE 8.3

- Express the trigonometric ratios $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$.
- Write all the other trigonometric ratios of $\angle A$ in terms of $\sec A$.
- Choose the correct option. Justify your choice.
 - $9 \sec^2 A - 9 \tan^2 A =$
 (A) 1 (B) 9 (C) 8 (D) 0
 - $(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta) =$
 (A) 0 (B) 1 (C) 2 (D) -1
 - $(\sec A + \tan A)(1 - \sin A) =$
 (A) $\sec A$ (B) $\sin A$ (C) $\operatorname{cosec} A$ (D) $\cos A$
 - $\frac{1 + \tan^2 A}{1 + \cot^2 A} =$
 (A) $\sec^2 A$ (B) -1 (C) $\cot^2 A$ (D) $\tan^2 A$
- Prove the following identities, where the angles involved are acute angles for which the expressions are defined.
 - $(\operatorname{cosec} \theta - \cot \theta)^2 = \frac{1 - \cos \theta}{1 + \cos \theta}$
 - $\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} = 2 \sec A$
 - $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$
[Hint : Write the expression in terms of $\sin \theta$ and $\cos \theta$]
 - $\frac{1 + \sec A}{\sec A} = \frac{\sin^2 A}{1 - \cos A}$ **[Hint : Simplify LHS and RHS separately]**
 - $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \operatorname{cosec} A + \cot A$, using the identity $\operatorname{cosec}^2 A = 1 + \cot^2 A$.
 - $\sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sec A + \tan A$
 - $\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} = \tan \theta$
 - $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$

$$(ix) (\operatorname{cosec} A - \sin A)(\sec A - \cos A) = \frac{1}{\tan A + \cot A}$$

[Hint : Simplify LHS and RHS separately]

$$(x) \left(\frac{1 + \tan^2 A}{1 + \cot^2 A} \right) = \left(\frac{1 - \tan A}{1 - \cot A} \right)^2 = \tan^2 A$$

8.5 Summary

In this chapter, you have studied the following points :

1. In a right triangle ABC, right-angled at B,
 $\sin A = \frac{\text{side opposite to angle } A}{\text{hypotenuse}}, \cos A = \frac{\text{side adjacent to angle } A}{\text{hypotenuse}}$
 $\tan A = \frac{\text{side opposite to angle } A}{\text{side adjacent to angle } A}$
2. $\operatorname{cosec} A = \frac{1}{\sin A}; \sec A = \frac{1}{\cos A}; \tan A = \frac{1}{\cot A}, \cot A = \frac{\sin A}{\cos A}$.
3. If one of the trigonometric ratios of an acute angle is known, the remaining trigonometric ratios of the angle can be easily determined.
4. The values of trigonometric ratios for angles $0^\circ, 30^\circ, 45^\circ, 60^\circ$ and 90° .
5. The value of $\sin A$ or $\cos A$ never exceeds 1, whereas the value of $\sec A$ ($0^\circ \leq A < 90^\circ$) or $\operatorname{cosec} A$ ($0^\circ < A \leq 90^\circ$) is always greater than or equal to 1.
6. $\sin^2 A + \cos^2 A = 1$,
 $\sec^2 A - \tan^2 A = 1$ for $0^\circ \leq A < 90^\circ$,
 $\operatorname{cosec}^2 A = 1 + \cot^2 A$ for $0^\circ < A \leq 90^\circ$.



1062CH04

QUADRATIC EQUATIONS

4

4.1 Introduction

In Chapter 2, you have studied different types of polynomials. One type was the quadratic polynomial of the form $ax^2 + bx + c$, $a \neq 0$. When we equate this polynomial to zero, we get a quadratic equation. Quadratic equations come up when we deal with many real-life situations. For instance, suppose a charity trust decides to build a prayer hall having a carpet area of 300 square metres with its length one metre more than twice its breadth. What should be the length and breadth of the hall? Suppose the breadth of the hall is x metres. Then, its length should be $(2x + 1)$ metres. We can depict this information pictorially as shown in Fig. 4.1.

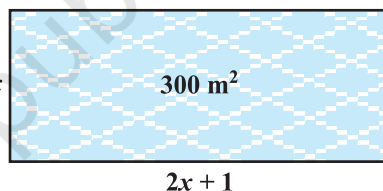


Fig. 4.1

Now, area of the hall = $(2x + 1) \cdot x \text{ m}^2 = (2x^2 + x) \text{ m}^2$

So, $2x^2 + x = 300$ (Given)

Therefore, $2x^2 + x - 300 = 0$

So, the breadth of the hall should satisfy the equation $2x^2 + x - 300 = 0$ which is a quadratic equation.

Many people believe that Babylonians were the first to solve quadratic equations. For instance, they knew how to find two positive numbers with a given positive sum and a given positive product, and this problem is equivalent to solving a quadratic equation of the form $x^2 - px + q = 0$. Greek mathematician Euclid developed a geometrical approach for finding out lengths which, in our present day terminology, are solutions of quadratic equations. Solving of quadratic equations, in general form, is often credited to ancient Indian mathematicians. In fact, Brahmagupta (C.E.598–665) gave an explicit formula to solve a quadratic equation of the form $ax^2 + bx = c$. Later,

Sridharacharya (C.E. 1025) derived a formula, now known as the quadratic formula, (as quoted by Bhaskara II) for solving a quadratic equation by the method of completing the square. An Arab mathematician Al-Khwarizmi (about C.E. 800) also studied quadratic equations of different types. Abraham bar Hiyya Ha-Nasi, in his book 'Liber embadorum' published in Europe in C.E. 1145 gave complete solutions of different quadratic equations.

In this chapter, you will study quadratic equations, and various ways of finding their roots. You will also see some applications of quadratic equations in daily life situations.

4.2 Quadratic Equations

A quadratic equation in the variable x is an equation of the form $ax^2 + bx + c = 0$, where a, b, c are real numbers, $a \neq 0$. For example, $2x^2 + x - 300 = 0$ is a quadratic equation. Similarly, $2x^2 - 3x + 1 = 0$, $4x - 3x^2 + 2 = 0$ and $1 - x^2 + 300 = 0$ are also quadratic equations.

In fact, any equation of the form $p(x) = 0$, where $p(x)$ is a polynomial of degree 2, is a quadratic equation. But when we write the terms of $p(x)$ in descending order of their degrees, then we get the standard form of the equation. That is, $ax^2 + bx + c = 0$, $a \neq 0$ is called the **standard form of a quadratic equation**.

Quadratic equations arise in several situations in the world around us and in different fields of mathematics. Let us consider a few examples.

Example 1 : Represent the following situations mathematically:

- John and Jivanti together have 45 marbles. Both of them lost 5 marbles each, and the product of the number of marbles they now have is 124. We would like to find out how many marbles they had to start with.
- A cottage industry produces a certain number of toys in a day. The cost of production of each toy (in rupees) was found to be 55 minus the number of toys produced in a day. On a particular day, the total cost of production was ₹ 750. We would like to find out the number of toys produced on that day.

Solution :

- Let the number of marbles John had be x .

Then the number of marbles Jivanti had = $45 - x$ (Why?).

The number of marbles left with John, when he lost 5 marbles = $x - 5$

The number of marbles left with Jivanti, when she lost 5 marbles = $45 - x - 5$
= $40 - x$

$$\begin{aligned}
 \text{Therefore, their product} &= (x - 5)(40 - x) \\
 &= 40x - x^2 - 200 + 5x \\
 &= -x^2 + 45x - 200
 \end{aligned}$$

$$\text{So, } -x^2 + 45x - 200 = 124 \quad (\text{Given that product} = 124)$$

$$\text{i.e., } -x^2 + 45x - 324 = 0$$

$$\text{i.e., } x^2 - 45x + 324 = 0$$

Therefore, the number of marbles John had, satisfies the quadratic equation

$$x^2 - 45x + 324 = 0$$

which is the required representation of the problem mathematically.

(ii) Let the number of toys produced on that day be x .

Therefore, the cost of production (in rupees) of each toy that day = $55 - x$

So, the total cost of production (in rupees) that day = $x(55 - x)$

$$\text{Therefore, } x(55 - x) = 750$$

$$\text{i.e., } 55x - x^2 = 750$$

$$\text{i.e., } -x^2 + 55x - 750 = 0$$

$$\text{i.e., } x^2 - 55x + 750 = 0$$

Therefore, the number of toys produced that day satisfies the quadratic equation

$$x^2 - 55x + 750 = 0$$

which is the required representation of the problem mathematically.

Example 2 : Check whether the following are quadratic equations:

$$(i) (x - 2)^2 + 1 = 2x - 3$$

$$(ii) x(x + 1) + 8 = (x + 2)(x - 2)$$

$$(iii) x(2x + 3) = x^2 + 1$$

$$(iv) (x + 2)^3 = x^3 - 4$$

Solution :

$$(i) \text{ LHS} = (x - 2)^2 + 1 = x^2 - 4x + 4 + 1 = x^2 - 4x + 5$$

Therefore, $(x - 2)^2 + 1 = 2x - 3$ can be rewritten as

$$x^2 - 4x + 5 = 2x - 3$$

$$\text{i.e., } x^2 - 6x + 8 = 0$$

It is of the form $ax^2 + bx + c = 0$.

Therefore, the given equation is a quadratic equation.

- (ii) Since $x(x + 1) + 8 = x^2 + x + 8$ and $(x + 2)(x - 2) = x^2 - 4$

Therefore, $x^2 + x + 8 = x^2 - 4$

i.e., $x + 12 = 0$

It is not of the form $ax^2 + bx + c = 0$.

Therefore, the given equation is not a quadratic equation.

- (iii) Here, LHS = $x(2x + 3) = 2x^2 + 3x$

So, $x(2x + 3) = x^2 + 1$ can be rewritten as

$$2x^2 + 3x = x^2 + 1$$

Therefore, we get $x^2 + 3x - 1 = 0$

It is of the form $ax^2 + bx + c = 0$.

So, the given equation is a quadratic equation.

- (iv) Here, LHS = $(x + 2)^3 = x^3 + 6x^2 + 12x + 8$

Therefore, $(x + 2)^3 = x^3 - 4$ can be rewritten as

$$x^3 + 6x^2 + 12x + 8 = x^3 - 4$$

i.e., $6x^2 + 12x + 12 = 0$ or, $x^2 + 2x + 2 = 0$

It is of the form $ax^2 + bx + c = 0$.

So, the given equation is a quadratic equation.

Remark : Be careful! In (ii) above, the given equation appears to be a quadratic equation, but it is not a quadratic equation.

In (iv) above, the given equation appears to be a cubic equation (an equation of degree 3) and not a quadratic equation. But it turns out to be a quadratic equation. As you can see, often we need to simplify the given equation before deciding whether it is quadratic or not.

EXERCISE 4.1

1. Check whether the following are quadratic equations :

(i) $(x + 1)^2 = 2(x - 3)$

(ii) $x^2 - 2x = (-2)(3 - x)$

(iii) $(x - 2)(x + 1) = (x - 1)(x + 3)$

(iv) $(x - 3)(2x + 1) = x(x + 5)$

(v) $(2x - 1)(x - 3) = (x + 5)(x - 1)$

(vi) $x^2 + 3x + 1 = (x - 2)^2$

(vii) $(x + 2)^3 = 2x(x^2 - 1)$

(viii) $x^3 - 4x^2 - x + 1 = (x - 2)^3$

2. Represent the following situations in the form of quadratic equations :

- (i) The area of a rectangular plot is 528 m². The length of the plot (in metres) is one more than twice its breadth. We need to find the length and breadth of the plot.

- (ii) The product of two consecutive positive integers is 306. We need to find the integers.
- (iii) Rohan's mother is 26 years older than him. The product of their ages (in years) 3 years from now will be 360. We would like to find Rohan's present age.
- (iv) A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. We need to find the speed of the train.

4.3 Solution of a Quadratic Equation by Factorisation

Consider the quadratic equation $2x^2 - 3x + 1 = 0$. If we replace x by 1 on the LHS of this equation, we get $(2 \times 1^2) - (3 \times 1) + 1 = 0 = \text{RHS}$ of the equation. We say that 1 is a root of the quadratic equation $2x^2 - 3x + 1 = 0$. This also means that 1 is a zero of the quadratic polynomial $2x^2 - 3x + 1$.

In general, a real number α is called a root of the quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$ if $a\alpha^2 + b\alpha + c = 0$. We also say that $x = \alpha$ is a **solution of the quadratic equation**, or that α **satisfies the quadratic equation**. Note that **the zeroes of the quadratic polynomial $ax^2 + bx + c$ and the roots of the quadratic equation $ax^2 + bx + c = 0$ are the same**.

You have observed, in Chapter 2, that a quadratic polynomial can have at most two zeroes. So, any quadratic equation can have at most two roots.

You have learnt in Class IX, how to factorise quadratic polynomials by splitting their middle terms. We shall use this knowledge for finding the roots of a quadratic equation. Let us see how.

Example 3 : Find the roots of the equation $2x^2 - 5x + 3 = 0$, by factorisation.

Solution : Let us first split the middle term $-5x$ as $-2x - 3x$ [because $(-2x) \times (-3x) = 6x^2 = (2x^2) \times 3$].

$$\text{So, } 2x^2 - 5x + 3 = 2x^2 - 2x - 3x + 3 = 2x(x - 1) - 3(x - 1) = (2x - 3)(x - 1)$$

Now, $2x^2 - 5x + 3 = 0$ can be rewritten as $(2x - 3)(x - 1) = 0$.

So, the values of x for which $2x^2 - 5x + 3 = 0$ are the same for which $(2x - 3)(x - 1) = 0$, i.e., either $2x - 3 = 0$ or $x - 1 = 0$.

Now, $2x - 3 = 0$ gives $x = \frac{3}{2}$ and $x - 1 = 0$ gives $x = 1$.

So, $x = \frac{3}{2}$ and $x = 1$ are the solutions of the equation.

In other words, 1 and $\frac{3}{2}$ are the roots of the equation $2x^2 - 5x + 3 = 0$.

Verify that these are the roots of the given equation.

Note that we have found the roots of $2x^2 - 5x + 3 = 0$ by factorising $2x^2 - 5x + 3$ into two linear factors and equating each factor to zero.

Example 4 : Find the roots of the quadratic equation $6x^2 - x - 2 = 0$.

Solution : We have

$$\begin{aligned} 6x^2 - x - 2 &= 6x^2 + 3x - 4x - 2 \\ &= 3x(2x + 1) - 2(2x + 1) \\ &= (3x - 2)(2x + 1) \end{aligned}$$

The roots of $6x^2 - x - 2 = 0$ are the values of x for which $(3x - 2)(2x + 1) = 0$.
Therefore, $3x - 2 = 0$ or $2x + 1 = 0$,

i.e.,
$$x = \frac{2}{3} \quad \text{or} \quad x = -\frac{1}{2}$$

Therefore, the roots of $6x^2 - x - 2 = 0$ are $\frac{2}{3}$ and $-\frac{1}{2}$.

We verify the roots, by checking that $\frac{2}{3}$ and $-\frac{1}{2}$ satisfy $6x^2 - x - 2 = 0$.

Example 5 : Find the roots of the quadratic equation $3x^2 - 2\sqrt{6}x + 2 = 0$.

Solution :

$$\begin{aligned} 3x^2 - 2\sqrt{6}x + 2 &= 3x^2 - \sqrt{6}x - \sqrt{6}x + 2 \\ &= \sqrt{3}x(\sqrt{3}x - \sqrt{2}) - \sqrt{2}(\sqrt{3}x - \sqrt{2}) \\ &= (\sqrt{3}x - \sqrt{2})(\sqrt{3}x - \sqrt{2}) \end{aligned}$$

So, the roots of the equation are the values of x for which

$$(\sqrt{3}x - \sqrt{2})(\sqrt{3}x - \sqrt{2}) = 0$$

Now, $\sqrt{3}x - \sqrt{2} = 0$ for $x = \sqrt{\frac{2}{3}}$.

So, this root is repeated twice, one for each repeated factor $\sqrt{3}x - \sqrt{2}$.

Therefore, the roots of $3x^2 - 2\sqrt{6}x + 2 = 0$ are $\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}$.

Example 6 : Find the dimensions of the prayer hall discussed in Section 4.1.

Solution : In Section 4.1, we found that if the breadth of the hall is x m, then x satisfies the equation $2x^2 + x - 300 = 0$. Applying the factorisation method, we write this equation as

$$2x^2 - 24x + 25x - 300 = 0$$

$$2x(x - 12) + 25(x - 12) = 0$$

i.e., $(x - 12)(2x + 25) = 0$

So, the roots of the given equation are $x = 12$ or $x = -12.5$. Since x is the breadth of the hall, it cannot be negative.

Thus, the breadth of the hall is 12 m. Its length = $2x + 1 = 25$ m.

EXERCISE 4.2

1. Find the roots of the following quadratic equations by factorisation:

(i) $x^2 - 3x - 10 = 0$

(ii) $2x^2 + x - 6 = 0$

(iii) $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$

(iv) $2x^2 - x + \frac{1}{8} = 0$

(v) $100x^2 - 20x + 1 = 0$

2. Solve the problems given in Example 1.

3. Find two numbers whose sum is 27 and product is 182.

4. Find two consecutive positive integers, sum of whose squares is 365.

5. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.

6. A cottage industry produces a certain number of pottery articles in a day. It was observed on a particular day that the cost of production of each article (in rupees) was 3 more than twice the number of articles produced on that day. If the total cost of production on that day was ₹ 90, find the number of articles produced and the cost of each article.

4.4 Nature of Roots

The equation $ax^2 + bx + c = 0$ are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If $b^2 - 4ac > 0$, we get two distinct real roots $-\frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a}$ and

$$-\frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a}.$$

If $b^2 - 4ac = 0$, then $x = -\frac{b}{2a} \pm 0$, i.e., $x = -\frac{b}{2a}$ or $-\frac{b}{2a}$.

So, the roots of the equation $ax^2 + bx + c = 0$ are both $-\frac{b}{2a}$.

Therefore, we say that the quadratic equation $ax^2 + bx + c = 0$ has two equal real roots in this case.

If $b^2 - 4ac < 0$, then there is no real number whose square is $b^2 - 4ac$. Therefore, there are no real roots for the given quadratic equation in this case.

Since $b^2 - 4ac$ determines whether the quadratic equation $ax^2 + bx + c = 0$ has real roots or not, $b^2 - 4ac$ is called the **discriminant** of this quadratic equation.

So, a quadratic equation $ax^2 + bx + c = 0$ has

- (i) **two distinct real roots, if $b^2 - 4ac > 0$,**
- (ii) **two equal real roots, if $b^2 - 4ac = 0$,**
- (iii) **no real roots, if $b^2 - 4ac < 0$.**

Let us consider some examples.

Example 7: Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$, and hence find the nature of its roots.

Solution : The given equation is of the form $ax^2 + bx + c = 0$, where $a = 2$, $b = -4$ and $c = 3$. Therefore, the discriminant

$$b^2 - 4ac = (-4)^2 - (4 \times 2 \times 3) = 16 - 24 = -8 < 0$$

So, the given equation has no real roots.

Example 8 : A pole has to be erected at a point on the boundary of a circular park of diameter 13 metres in such a way that the differences of its distances from two diametrically opposite fixed gates A and B on the boundary is 7 metres. Is it possible to do so? If yes, at what distances from the two gates should the pole be erected?

Solution : Let us first draw the diagram (see Fig. 4.2).

Let P be the required location of the pole. Let the distance of the pole from the gate B be x m, i.e., $BP = x$ m. Now the difference of the distances of the pole from the two gates = $AP - BP$ (or, $BP - AP$) = 7 m. Therefore, $AP = (x + 7)$ m.

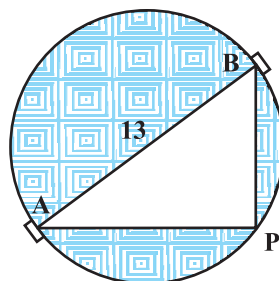


Fig. 4.2

Now, $AB = 13\text{m}$, and since AB is a diameter,

$$\angle APB = 90^\circ \quad (\text{Why?})$$

Therefore, $AP^2 + PB^2 = AB^2$ (By Pythagoras theorem)

$$\text{i.e.,} \quad (x + 7)^2 + x^2 = 13^2$$

$$\text{i.e.,} \quad x^2 + 14x + 49 + x^2 = 169$$

$$\text{i.e.,} \quad 2x^2 + 14x - 120 = 0$$

So, the distance 'x' of the pole from gate B satisfies the equation

$$x^2 + 7x - 60 = 0$$

So, it would be possible to place the pole if this equation has real roots. To see if this is so or not, let us consider its discriminant. The discriminant is

$$b^2 - 4ac = 7^2 - 4 \times 1 \times (-60) = 289 > 0.$$

So, the given quadratic equation has two real roots, and it is possible to erect the pole on the boundary of the park.

Solving the quadratic equation $x^2 + 7x - 60 = 0$, by the quadratic formula, we get

$$x = \frac{-7 \pm \sqrt{289}}{2} = \frac{-7 \pm 17}{2}$$

Therefore, $x = 5$ or -12 .

Since x is the distance between the pole and the gate B, it must be positive. Therefore, $x = -12$ will have to be ignored. So, $x = 5$.

Thus, the pole has to be erected on the boundary of the park at a distance of 5m from the gate B and 12m from the gate A.

Example 9 : Find the discriminant of the equation $3x^2 - 2x + \frac{1}{3} = 0$ and hence find the nature of its roots. Find them, if they are real.

Solution : Here $a = 3$, $b = -2$ and $c = \frac{1}{3}$.

Therefore, discriminant $b^2 - 4ac = (-2)^2 - 4 \times 3 \times \frac{1}{3} = 4 - 4 = 0$.

Hence, the given quadratic equation has two equal real roots.

The roots are $\frac{-b}{2a}, \frac{-b}{2a}$, i.e., $\frac{2}{6}, \frac{2}{6}$, i.e., $\frac{1}{3}, \frac{1}{3}$.

EXERCISE 4.3

- Find the nature of the roots of the following quadratic equations. If the real roots exist, find them:
 - $2x^2 - 3x + 5 = 0$
 - $3x^2 - 4\sqrt{3}x + 4 = 0$
 - $2x^2 - 6x + 3 = 0$
- Find the values of k for each of the following quadratic equations, so that they have two equal roots.
 - $2x^2 + kx + 3 = 0$
 - $kx(x - 2) + 6 = 0$
- Is it possible to design a rectangular mango grove whose length is twice its breadth, and the area is 800 m^2 ? If so, find its length and breadth.
- Is the following situation possible? If so, determine their present ages.
The sum of the ages of two friends is 20 years. Four years ago, the product of their ages in years was 48.
- Is it possible to design a rectangular park of perimeter 80 m and area 400 m^2 ? If so, find its length and breadth.

4.5 Summary

In this chapter, you have studied the following points:

- A quadratic equation in the variable x is of the form $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$.
- A real number α is said to be a root of the quadratic equation $ax^2 + bx + c = 0$, if $a\alpha^2 + b\alpha + c = 0$. The zeroes of the quadratic polynomial $ax^2 + bx + c$ and the roots of the quadratic equation $ax^2 + bx + c = 0$ are the same.
- If we can factorise $ax^2 + bx + c$, $a \neq 0$, into a product of two linear factors, then the roots of the quadratic equation $ax^2 + bx + c = 0$ can be found by equating each factor to zero.
- Quadratic formula: The roots of a quadratic equation $ax^2 + bx + c = 0$ are given by
$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \text{ provided } b^2 - 4ac \geq 0.$$
- A quadratic equation $ax^2 + bx + c = 0$ has
 - two distinct real roots, if $b^2 - 4ac > 0$,
 - two equal roots (i.e., coincident roots), if $b^2 - 4ac = 0$, and
 - no real roots, if $b^2 - 4ac < 0$.

NOTE

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1062CH01

REAL NUMBERS

1

1.1 Introduction

In Class IX, you began your exploration of the world of real numbers and encountered irrational numbers. We continue our discussion on real numbers in this chapter. We begin with very important properties of positive integers in Sections 1.2, namely the Euclid's division algorithm and the Fundamental Theorem of Arithmetic.

Euclid's division algorithm, as the name suggests, has to do with divisibility of integers. Stated simply, it says any positive integer a can be divided by another positive integer b in such a way that it leaves a remainder r that is smaller than b . Many of you probably recognise this as the usual long division process. Although this result is quite easy to state and understand, it has many applications related to the divisibility properties of integers. We touch upon a few of them, and use it mainly to compute the HCF of two positive integers.

The Fundamental Theorem of Arithmetic, on the other hand, has to do something with multiplication of positive integers. You already know that every composite number can be expressed as a product of primes in a unique way—this important fact is the Fundamental Theorem of Arithmetic. Again, while it is a result that is easy to state and understand, it has some very deep and significant applications in the field of mathematics. We use the Fundamental Theorem of Arithmetic for two main applications. First, we use it to prove the irrationality of many of the numbers you studied in Class IX, such as $\sqrt{2}$, $\sqrt{3}$ and $\sqrt{5}$. Second, we apply this theorem to explore when exactly the decimal expansion of a rational number, say $\frac{p}{q}$ ($q \neq 0$), is terminating and when it is non-terminating repeating. We do so by looking at the prime factorisation of the denominator q of $\frac{p}{q}$. You will see that the prime factorisation of q will completely reveal the nature of the decimal expansion of $\frac{p}{q}$.

So let us begin our exploration.

1.2 The Fundamental Theorem of Arithmetic

In your earlier classes, you have seen that any natural number can be written as a product of its prime factors. For instance, $2 = 2$, $4 = 2 \times 2$, $253 = 11 \times 23$, and so on. Now, let us try and look at natural numbers from the other direction. That is, can any natural number be obtained by multiplying prime numbers? Let us see.

Take any collection of prime numbers, say 2, 3, 7, 11 and 23. If we multiply some or all of these numbers, allowing them to repeat as many times as we wish, we can produce a large collection of positive integers (In fact, infinitely many). Let us list a few :

$$7 \times 11 \times 23 = 1771$$

$$3 \times 7 \times 11 \times 23 = 5313$$

$$2 \times 3 \times 7 \times 11 \times 23 = 10626$$

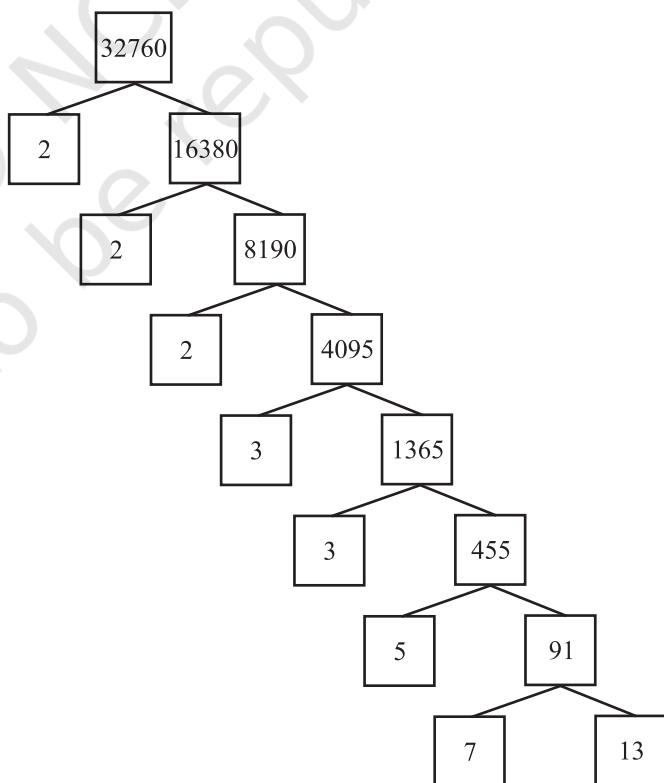
$$2^3 \times 3 \times 7^3 = 8232$$

$$2^2 \times 3 \times 7 \times 11 \times 23 = 21252$$

and so on.

Now, let us suppose your collection of primes includes all the possible primes. What is your guess about the size of this collection? Does it contain only a finite number of integers, or infinitely many? Infact, there are infinitely many primes. So, if we combine all these primes in all possible ways, we will get an infinite collection of numbers, all the primes and all possible products of primes. The question is – can we produce all the composite numbers this way? What do you think? Do you think that there may be a composite number which is not the product of powers of primes? Before we answer this, let us factorise positive integers, that is, do the opposite of what we have done so far.

We are going to use the factor tree with which you are all familiar. Let us take some large number, say, 32760, and factorise it as shown.



So we have factorised 32760 as $2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 \times 13$ as a product of primes, i.e., $32760 = 2^3 \times 3^2 \times 5 \times 7 \times 13$ as a product of powers of primes. Let us try another number, say, 123456789. This can be written as $3^2 \times 3803 \times 3607$. Of course, you have to check that 3803 and 3607 are primes! (Try it out for several other natural numbers yourself.) This leads us to a conjecture that every composite number can be written as the product of powers of primes. In fact, this statement is true, and is called the **Fundamental Theorem of Arithmetic** because of its basic crucial importance to the study of integers. Let us now formally state this theorem.

Theorem 1.1 (Fundamental Theorem of Arithmetic) : *Every composite number can be expressed (factorised) as a product of primes, and this factorisation is unique, apart from the order in which the prime factors occur.*

An equivalent version of Theorem 1.2 was probably first recorded as Proposition 14 of Book IX in Euclid's Elements, before it came to be known as the Fundamental Theorem of Arithmetic. However, the first correct proof was given by Carl Friedrich Gauss in his *Disquisitiones Arithmeticae*.

Carl Friedrich Gauss is often referred to as the 'Prince of Mathematicians' and is considered one of the three greatest mathematicians of all time, along with Archimedes and Newton. He has made fundamental contributions to both mathematics and science.



Carl Friedrich Gauss
(1777 – 1855)

The Fundamental Theorem of Arithmetic says that every composite number can be factorised as a product of primes. Actually it says more. It says that given any composite number it can be factorised as a product of prime numbers in a '**unique**' way, except for the order in which the primes occur. That is, given any composite number there is one and only one way to write it as a product of primes, as long as we are not particular about the order in which the primes occur. So, for example, we regard $2 \times 3 \times 5 \times 7$ as the same as $3 \times 5 \times 7 \times 2$, or any other possible order in which these primes are written. This fact is also stated in the following form:

The prime factorisation of a natural number is unique, except for the order of its factors.

In general, given a composite number x , we factorise it as $x = p_1 p_2 \dots p_n$, where $p_1, p_2, \dots, p_n$ are primes and written in ascending order, i.e., $p_1 \leq p_2 \leq \dots \leq p_n$. If we combine the same primes, we will get powers of primes. For example,

$$32760 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 \times 13 = 2^3 \times 3^2 \times 5 \times 7 \times 13$$

Once we have decided that the order will be ascending, then the way the number is factorised, is unique.

The Fundamental Theorem of Arithmetic has many applications, both within mathematics and in other fields. Let us look at some examples.

Example 1 : Consider the numbers 4^n , where n is a natural number. Check whether there is any value of n for which 4^n ends with the digit zero.

Solution : If the number 4^n , for any n , were to end with the digit zero, then it would be divisible by 5. That is, the prime factorisation of 4^n would contain the prime 5. This is not possible because $4^n = (2)^{2n}$; so the only prime in the factorisation of 4^n is 2. So, the uniqueness of the Fundamental Theorem of Arithmetic guarantees that there are no other primes in the factorisation of 4^n . So, there is no natural number n for which 4^n ends with the digit zero.

You have already learnt how to find the HCF and LCM of two positive integers using the Fundamental Theorem of Arithmetic in earlier classes, without realising it! This method is also called the *prime factorisation method*. Let us recall this method through an example.

Example 2 : Find the LCM and HCF of 6 and 20 by the prime factorisation method.

Solution : We have : $6 = 2^1 \times 3^1$ and $20 = 2 \times 2 \times 5 = 2^2 \times 5^1$.

You can find $\text{HCF}(6, 20) = 2$ and $\text{LCM}(6, 20) = 2 \times 2 \times 3 \times 5 = 60$, as done in your earlier classes.

Note that $\text{HCF}(6, 20) = 2^1 =$ **Product of the smallest power of each common prime factor in the numbers.**

$\text{LCM}(6, 20) = 2^2 \times 3^1 \times 5^1 =$ **Product of the greatest power of each prime factor, involved in the numbers.**

From the example above, you might have noticed that $\text{HCF}(6, 20) \times \text{LCM}(6, 20) = 6 \times 20$. In fact, we can verify that **for any two positive integers a and b , $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$** . We can use this result to find the LCM of two positive integers, if we have already found the HCF of the two positive integers.

Example 3: Find the HCF of 96 and 404 by the prime factorisation method. Hence, find their LCM.

Solution : The prime factorisation of 96 and 404 gives :

$$96 = 2^5 \times 3, 404 = 2^2 \times 101$$

Therefore, the HCF of these two integers is $2^2 = 4$.

Also,

$$\text{LCM}(96, 404) = \frac{96 \times 404}{\text{HCF}(96, 404)} = \frac{96 \times 404}{4} = 9696$$

Example 4 : Find the HCF and LCM of 6, 72 and 120, using the prime factorisation method.

Solution : We have :

$$6 = 2 \times 3, 72 = 2^3 \times 3^2, 120 = 2^3 \times 3 \times 5$$

Here, 2^1 and 3^1 are the smallest powers of the common factors 2 and 3, respectively.

So,

$$\text{HCF}(6, 72, 120) = 2^1 \times 3^1 = 2 \times 3 = 6$$

2^3 , 3^2 and 5^1 are the greatest powers of the prime factors 2, 3 and 5 respectively involved in the three numbers.

So,

$$\text{LCM}(6, 72, 120) = 2^3 \times 3^2 \times 5^1 = 360$$

Remark : Notice, $6 \times 72 \times 120 \neq \text{HCF}(6, 72, 120) \times \text{LCM}(6, 72, 120)$. So, the product of three numbers is not equal to the product of their HCF and LCM.

EXERCISE 1.1

- Express each number as a product of its prime factors:
(i) 140 (ii) 156 (iii) 3825 (iv) 5005 (v) 7429
- Find the LCM and HCF of the following pairs of integers and verify that $\text{LCM} \times \text{HCF} =$ product of the two numbers.
(i) 26 and 91 (ii) 510 and 92 (iii) 336 and 54
- Find the LCM and HCF of the following integers by applying the prime factorisation method.
(i) 12, 15 and 21 (ii) 17, 23 and 29 (iii) 8, 9 and 25
- Given that $\text{HCF}(306, 657) = 9$, find $\text{LCM}(306, 657)$.
- Check whether 6^n can end with the digit 0 for any natural number n .
- Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers.
- There is a circular path around a sports field. Sonia takes 18 minutes to drive one round of the field, while Ravi takes 12 minutes for the same. Suppose they both start at the

same point and at the same time, and go in the same direction. After how many minutes will they meet again at the starting point?

1.3 Revisiting Irrational Numbers

In Class IX, you were introduced to irrational numbers and many of their properties. You studied about their existence and how the rationals and the irrationals together made up the real numbers. You even studied how to locate irrationals on the number line. However, we did not prove that they were irrationals. In this section, we will prove that $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ and, in general, $\sqrt{p}$ is irrational, where p is a prime. One of the theorems, we use in our proof, is the Fundamental Theorem of Arithmetic.

Recall, a number 's' is called *irrational* if it cannot be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$. Some examples of irrational numbers, with which you are already familiar, are :

$$\sqrt{2}, \sqrt{3}, \sqrt{15}, \pi, -\frac{\sqrt{2}}{\sqrt{3}}, 0.10110111011110\dots, \text{ etc.}$$

Before we prove that $\sqrt{2}$ is irrational, we need the following theorem, whose proof is based on the Fundamental Theorem of Arithmetic.

Theorem 1.2 : Let p be a prime number. If p divides a^2 , then p divides a , where a is a positive integer.

***Proof :** Let the prime factorisation of a be as follows :

$$a = p_1 p_2 \dots p_n, \text{ where } p_1, p_2, \dots, p_n \text{ are primes, not necessarily distinct.}$$

$$\text{Therefore, } a^2 = (p_1 p_2 \dots p_n)(p_1 p_2 \dots p_n) = p_1^2 p_2^2 \dots p_n^2.$$

Now, we are given that p divides a^2 . Therefore, from the Fundamental Theorem of Arithmetic, it follows that p is one of the prime factors of a^2 . However, using the uniqueness part of the Fundamental Theorem of Arithmetic, we realise that the only prime factors of a^2 are $p_1, p_2, \dots, p_n$. So p is one of $p_1, p_2, \dots, p_n$.

Now, since $a = p_1 p_2 \dots p_n$, p divides a .

We are now ready to give a proof that $\sqrt{2}$ is irrational.

The proof is based on a technique called 'proof by contradiction'. (This technique is discussed in some detail in Appendix 1).

Theorem 1.3 : $\sqrt{2}$ is irrational.

Proof : Let us assume, to the contrary, that $\sqrt{2}$ is rational.

* Not from the examination point of view.

So, we can find integers r and s ($\neq 0$) such that $\sqrt{2} = \frac{r}{s}$.

Suppose r and s have a common factor other than 1. Then, we divide by the common factor to get $\sqrt{2} = \frac{a}{b}$, where a and b are coprime.

So, $b\sqrt{2} = a$.

Squaring on both sides and rearranging, we get $2b^2 = a^2$. Therefore, 2 divides a^2 .

Now, by Theorem 1.2, it follows that 2 divides a .

So, we can write $a = 2c$ for some integer c .

Substituting for a , we get $2b^2 = 4c^2$, that is, $b^2 = 2c^2$.

This means that 2 divides b^2 , and so 2 divides b (again using Theorem 1.2 with $p = 2$).

Therefore, a and b have at least 2 as a common factor.

But this contradicts the fact that a and b have no common factors other than 1.

This contradiction has arisen because of our incorrect assumption that $\sqrt{2}$ is rational.

So, we conclude that $\sqrt{2}$ is irrational.

Example 5 : Prove that $\sqrt{3}$ is irrational.

Solution : Let us assume, to the contrary, that $\sqrt{3}$ is rational.

That is, we can find integers a and b ($\neq 0$) such that $\sqrt{3} = \frac{a}{b}$.

Suppose a and b have a common factor other than 1, then we can divide by the common factor, and assume that a and b are coprime.

So, $b\sqrt{3} = a$.

Squaring on both sides, and rearranging, we get $3b^2 = a^2$.

Therefore, a^2 is divisible by 3, and by Theorem 1.2, it follows that a is also divisible by 3.

So, we can write $a = 3c$ for some integer c .

Substituting for a , we get $3b^2 = 9c^2$, that is, $b^2 = 3c^2$.

This means that b^2 is divisible by 3, and so b is also divisible by 3 (using Theorem 1.2 with $p = 3$).

Therefore, a and b have at least 3 as a common factor.

But this contradicts the fact that a and b are coprime.

This contradiction has arisen because of our incorrect assumption that $\sqrt{3}$ is rational.

So, we conclude that $\sqrt{3}$ is irrational.

In Class IX, we mentioned that :

- the sum or difference of a rational and an irrational number is irrational and
- the product and quotient of a non-zero rational and irrational number is irrational.

We prove some particular cases here.

Example 6 : Show that $5 - \sqrt{3}$ is irrational.

Solution : Let us assume, to the contrary, that $5 - \sqrt{3}$ is rational.

That is, we can find coprime a and b ($b \neq 0$) such that $5 - \sqrt{3} = \frac{a}{b}$.

Therefore, $5 - \frac{a}{b} = \sqrt{3}$.

Rearranging this equation, we get $\sqrt{3} = 5 - \frac{a}{b} = \frac{5b - a}{b}$.

Since a and b are integers, we get $5 - \frac{a}{b}$ is rational, and so $\sqrt{3}$ is rational.

But this contradicts the fact that $\sqrt{3}$ is irrational.

This contradiction has arisen because of our incorrect assumption that $5 - \sqrt{3}$ is rational.

So, we conclude that $5 - \sqrt{3}$ is irrational.

Example 7 : Show that $3\sqrt{2}$ is irrational.

Solution : Let us assume, to the contrary, that $3\sqrt{2}$ is rational.

That is, we can find coprime a and b ($b \neq 0$) such that $3\sqrt{2} = \frac{a}{b}$.

Rearranging, we get $\sqrt{2} = \frac{a}{3b}$.

Since 3, a and b are integers, $\frac{a}{3b}$ is rational, and so $\sqrt{2}$ is rational.

But this contradicts the fact that $\sqrt{2}$ is irrational.

So, we conclude that $3\sqrt{2}$ is irrational.

EXERCISE 1.2

1. Prove that $\sqrt{5}$ is irrational.
2. Prove that $3 + 2\sqrt{5}$ is irrational.
3. Prove that the following are irrationals :

(i) $\frac{1}{\sqrt{2}}$

(ii) $7\sqrt{5}$

(iii) $6 + \sqrt{2}$

1.4 Summary

In this chapter, you have studied the following points:

1. The Fundamental Theorem of Arithmetic :

Every composite number can be expressed (factorised) as a product of primes, and this factorisation is unique, apart from the order in which the prime factors occur.

2. If p is a prime and p divides a^2 , then p divides a , where a is a positive integer.
3. To prove that $\sqrt{2}$, $\sqrt{3}$ are irrationals.

A NOTE TO THE READER

You have seen that :

$\text{HCF}(p, q, r) \times \text{LCM}(p, q, r) \neq p \times q \times r$, where p, q, r are positive integers (see Example 8). However, the following results hold good for three numbers p, q and r :

$$\text{LCM}(p, q, r) = \frac{p \cdot q \cdot r \cdot \text{HCF}(p, q, r)}{\text{HCF}(p, q) \cdot \text{HCF}(q, r) \cdot \text{HCF}(p, r)}$$

$$\text{HCF}(p, q, r) = \frac{p \cdot q \cdot r \cdot \text{LCM}(p, q, r)}{\text{LCM}(p, q) \cdot \text{LCM}(q, r) \cdot \text{LCM}(p, r)}$$